

ICA Lens: Interpreting Language Models Without Training Another Dictionary

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Finding interpretable directions in language-model representations is a critical step toward understanding model behavior and enabling more precise control over model internals. Sparse autoencoders (SAEs) have become the standard tool for this purpose, but using them as the default first lens often requires training, tuning, storing, and evaluating large overcomplete dictionaries for every layer, activation site, and sparsity setting. This computational bottleneck limits rapid exploration and motivates a fundamental question: *how much interpretable structure is already visible from activation geometry before training another neural dictionary?* Our intuition is simple: many interpretable directions are selective on tokens, and selective directions should look less Gaussian than typical random directions. We therefore revisit independent component analysis (ICA), a classical method for finding non-Gaussian directions, as a compact lens for language-model interpretability. We find that ICA has been underestimated for LLM interpretability because prior uses often relied on off-the-shelf ICA implementations that are brittle on LLM activations and lacked systematic tools for inspecting, annotating, and evaluating the recovered directions. To bridge these gaps, we introduce ICALENS, the first practical workflow for stable, efficient, and auditable ICA analysis of LLM representations. ICALENS combines an optimized GPU-parallel FastICA pipeline with LLM-specific stability recipes and better fitting diagnostics, enabling efficient and reliable layer-wise analysis across modern LLMs. Across GPT-2 Small, Gemma 2 2B, and Qwen 3.5 2B Base, ICALENS efficiently recovers compact, human-interpretable directions without per-layer gradient-based dictionary training. On SAE Bench, ICA is competitive with public SAEs in sparse probing and outperforms them in targeted probe perturbation under small-to-medium component budgets. These results suggest that ICA should not be viewed as a weak classical baseline, but as an efficient and complementary first lens for exploring language-model representations. To support reproducible analysis, we release all fitted ICA checkpoints, the ICA explorer, and human annotations.

Keywords: Mechanistic Interpretability, Independent Component Analysis, Sparse Autoencoders, Language Model Representations

 **Project Page:** [ICA Project Website](#)

 **Online Demo:** [ICA Explorer](#)

 **Checkpoints:** [Hugging Face Collection](#)

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1. Introduction

Understanding the internal representations of large language models (LLMs) is a central goal of mechanistic interpretability (Olah et al., 2020, Elhage et al., 2021, Gurnee et al., 2023, Park et al., 2023). Sparse autoencoders have become the dominant approach to this problem by decomposing activations into sparse latents over an overcomplete dictionary of learned directions (Huben et al., 2024, Bricken et al., 2023, Templeton et al., 2024, Gao et al., 2025). SAEs are powerful when a fine-grained feature dictionary is needed, but using them as the default first step comes with substantial practical costs. Training these dictionaries requires substantial data, storage, tuning, and compute. Gemma Scope, for example, required hundreds of SAEs, tens of millions of learned latents, 4–16B training tokens per SAE, roughly 20 PiB of saved activations, and over 20% of GPT-3’s training compute (Lieberum et al., 2024). This cost makes it difficult to train new dictionaries for every model, layer, activation site, and sparsity setting one wants to inspect. Publicly released SAEs partly mitigate the burden, but their coverage and release cadence remain limited compared with the pace and diversity of open model releases.

For many early-stage analyses, however, we may not need a full overcomplete dictionary. A compact set of interpretable directions can already help identify promising layers, inspect concepts, and guide lightweight interventions. This motivates a lighter first lens that can expose useful activation structure without training another large dictionary.

How much interpretable structure is already visible from activation geometry? To answer this question, we go back to a basic intuition about interpretability: many useful interpretable directions are selective. They should not behave like typical random projections over all tokens. Instead, they may activate on a small set of contexts, prefer one sign, or respond to lexical, syntactic, semantic, or discourse patterns. Such selectivity should leave a statistical footprint in activation space.

We study non-Gaussianity as a simple and measurable footprint of this kind. This is motivated by our observation that public SAE decoder directions are substantially more non-Gaussian than random directions, despite being optimized for sparse reconstruction rather than explicit non-Gaussianity. This suggests a simple possibility: by directly searching for highly non-Gaussian directions, we may recover part of the structure that SAEs learn through expensive dictionary training.

Independent component analysis (ICA) gives an intuitive way to test this possibility (Hyvarinen, 1999, Hyvarinen and Oja, 2000). After centering and whitening, ICA finds a linear basis whose one-dimensional projections are as non-Gaussian as possible, returning a small set of statistically exceptional directions. In this sense, ICA directly uses a signal that sparse reconstruction appears to learn only implicitly.

Despite its conceptual elegance, ICA is rarely used as a first lens for layer-wise LLM activation analysis. We argue that ICA has been underestimated for two practical reasons. First, standard ICA does not work reliably out of the box on modern LLM activations. Residual streams are high-dimensional, contain large-norm token outliers, and include a small number of slow or oscillating components that can dominate convergence. As a result, a naive FastICA implementation¹ can be slow, fail to converge on many layers, and return few or no usable components. Second, prior evaluations have not systematically tested the interpretability of ICA directions on LLM activations through human annotation or downstream measures of feature utility (Karvonen et al., 2025, Makelov et al., 2025). These implementation and evaluation gaps leave the practical value of ICA unclear.

¹For example, prior work used an unoptimized CPU FastICA pipeline from scikit-learn as its ICA baseline (Huben et al., 2024). See https://github.com/HoagyC/sparse_coding/blob/main/autoencoders/ica.py.

We revisit ICA for LLM interpretability and introduce ICALens, a practical workflow for stable, efficient, and auditable ICA analysis of LLM representations.

Making ICA practical for LLM activations (Section 3). We use three recipes to make fitting faster and more stable. Row normalization reduces the influence of activation-norm outliers before whitening. A p95-LIM rule accepts a fit when most components have stabilized but a small tail remains difficult to converge. Adaptive refitting lowers the target component count for difficult layers. On GPT-2 Small with 1M activations, our pipeline increases the number of accepted layers by 400.0% and reduces the total number of FastICA iterations by 21.5%.

Characterizing what ICA recovers (Section 4). ICA does not return an arbitrary rotation of the residual stream. Across GPT-2 Small, Gemma 2 2B, and Qwen 3.5 2B Base, its components are substantially more non-Gaussian than random directions and public SAE decoder directions. To study what textual evidence makes these directions exceptional, we introduce the effective receptive field (ERF), which measures how much textual evidence is needed for a direction to activate. ERF reveals a layer-wise shift from token-local components in shallow layers to broader context-dependent components in deeper layers. It also shows an empirical link between non-Gaussianity and interpretability—more non-Gaussian directions tend to require less context to activate, making them easier to explain from top examples.

Interpreting ICA components (Section 5). We build an interactive ICA explorer and manually annotate components across all three model families. Through random audits, secondary expert checks, controlled prompt tests, and case studies, we find that ICA components expose readable structure in both residual streams and token embeddings. The recovered components capture lexical form, local syntax, phrase templates, discourse context, semantic ambiguity, long-range repetition, and embedding-level lexical organization. These results show that ICA components are not merely high-kurtosis artifacts, but inspectable directions that support testable working hypotheses.

Testing feature utility (Section 6). Using SAE Bench Sparse Probing and Targeted Probe Perturbation (TPP), we test whether ICA directions both concentrate concept-relevant information and support selective interventions (Karvonen et al., 2025). We compare ICA with public high-capacity SAE dictionaries, Matryoshka SAE variants with smaller dictionary sizes (Bussmann et al., 2025), ITDA as a training-light sparse-coding alternative (Leask et al., 2025), and PCA as a classical statistical baseline (Abdi and Williams, 2010). Across models, ICA remains highly competitive with SAEs and consistently outperforms ITDA and PCA in sparse probing. It also achieves stronger TPP performance than public SAEs under small-to-medium intervention budgets. These results suggest that ICA directions are useful feature coordinates, and that non-Gaussianity is a more informative interpretability signal than variance.

Analyzing the ICA–SAE Relationship (Section 7). We find that non-Gaussianity maximization and sparse reconstruction recover related yet non-redundant directions. ICA recovers many SAE-aligned directions, while also exposing components that are weakly captured by any single SAE feature. We also find that the two methods exhibit distinct token-level activation patterns. ICA directions often show strong activations across a set of concept-related tokens, whereas SAE features more often activate on localized tokens. These findings position ICA as a complementary first lens for language-model activations, rather than as a replacement for sparse autoencoders. SAEs learn large dictionaries optimized for sparse reconstruction, making them well-suited for high-resolution feature discovery. ICA asks how much interpretable structure is already visible from activation geometry, helping us decide where heavier dictionary learning is worth the cost.

2. Related Work

Early work showed that sparse dictionary learning can recover latents that are more monosemantic, interpretable, and causally useful than neurons or classical decompositions (Huben et al., 2024, Bricken et al., 2023, Templeton et al., 2024, Gao et al., 2025). This success has made sparse autoencoders the dominant route for turning language-model activations into inspectable feature coordinates and has led to a broader ecosystem: automated feature labeling (Paulo et al., 2024), standardized evaluation (Karvonen et al., 2025, Makelov et al., 2025), improved architectures and objectives (e.g., Gated, TopK, JumpReLU, BatchTopK, Switch, and Matryoshka SAEs), and open model-wide releases such as Gemma Scope, Llama Scope, and Qwen Scope (Rajamanoharan et al., 2024a, Gao et al., 2025, Rajamanoharan et al., 2024b, Bussmann et al., 2024, Mudide et al., 2025, Bussmann et al., 2025, Lieberum et al., 2024, Deng et al., 2026).

Independent Component Analysis (ICA) is a classical lightweight alternative that searches for statistically independent, non-Gaussian directions in a linear representation space (Hyvarinen, 1999, Hyvarinen and Oja, 2000). This objective has recovered interpretable components in images, fMRI, word embeddings, and multimodal embedding spaces (Bell and Sejnowski, 1997, Daubechies et al., 2009, Yamagiwa et al., 2023, 2024, Musil and Mareček, 2024). However, ICA has received much less attention as a practical tool for analyzing modern LLM activations. Unlike SAEs, it has not been developed into a complete analysis workflow with stable fitting recipes, feature browsers, human annotation protocols, and task-level evaluations.

Our work revisits ICA for modern LLM interpretability. We build a more complete ICA workflow, including a more stable fitting pipeline, an annotation and analysis platform, and a broader evaluation suite. Our results show that ICA has been underestimated, and position ICA as a compact complement to SAEs: lower-capacity than an overcomplete dictionary, but cheaper to fit, easier to browse, and competitive with public SAEs while outperforming the lightweight baselines we evaluate.

3. ICALens: Making ICA Practical for LLM Activations

FastICA (Hyvarinen, 1999) is a fixed-point method for independent component analysis. Given cached activations $X^\ell \in \mathbb{R}^{n \times d}$ from layer ℓ , it first centers and whitens the activations to obtain $Z \in \mathbb{R}^{n \times m}$. After whitening removes second-order correlations, ICA searches for a rotation $W \in \mathbb{R}^{m \times m}$ whose coordinates are as non-Gaussian as possible. With the standard log-cosh contrast G , this can be viewed as

$$\max_{WW^\top = I} \frac{1}{n} \sum_{i=1}^n \sum_{j=1}^m G((ZW^\top)_{ij}), \quad S = ZW^\top,$$

where S contains the signed ICA component scores. In this work, we use the parallel FastICA variant and implement it in PyTorch so fitting can run efficiently on GPU.²

It is conceptually simple, but a direct application to modern LLM activations is brittle. Activation matrices are high-dimensional, often contain large-norm token positions, and can include a small number of slow or oscillating components that dominate worst-case convergence statistics. Thus, a naive FastICA run can reject an entire layer even when most directions have already stabilized and remain useful for analysis.

Algorithm 1 summarizes our layer-wise pipeline. We make FastICA practical with three simple recipes. First, row normalization reduces the influence of activation-norm outliers before whitening (Section 3.2). Second,

²We build on https://github.com/RichieHakim/FastICA_torch. The implementation follows the standard FastICA interface, similar in spirit to scikit-learn’s FastICA, while using PyTorch tensors and GPU-parallel linear algebra. We further modify it for numerical stability on LLM activations.

p95-LIM provides a fallback acceptance rule when most components have stabilized but a small tail remains difficult (Section 3.3). Third, adaptive refitting reduces the target component count only when needed, returning the highest accepted resolution for each layer under the same convergence standard (Section 3.4).

Algorithm 1 ICALens layer-wise ICA fitting pipeline. Highlighted steps are our additions to standard FastICA.

Require: Activation matrix X^ℓ , initial component count m_0 , LIM threshold τ , maximum iterations T

Ensure: Artifact $A_\ell = (\mu, K, W, R, D, S, I_{\text{st}}, I_{\text{tail}}, \text{status})$

```

1:  $M \leftarrow (m_0, \lfloor m_0/2 \rfloor, \lfloor m_0/4 \rfloor, \dots)$  [A3 adaptive component search]
2: for  $m \in M$  from largest to smallest do
3:   Normalize rows:  $\bar{x}_i^\ell \leftarrow x_i^\ell / \max(\|x_i^\ell\|_2, \epsilon)$ , for  $i = 1, \dots, n$  [A1 row normalization]
4:   Center activations:  $\mu \leftarrow n^{-1} \sum_i \bar{x}_i^\ell$ ,  $X_c^\ell \leftarrow \bar{X}^\ell - \mathbf{1}\mu^\top$ 
5:   Whiten activations: compute  $K$ ,  $Z \leftarrow X_c^\ell K^\top$ 
6:   Run parallel FastICA on  $Z$  with  $m$  components for at most  $T$  iterations
7:   Obtain rotation  $W$ , source scores  $S$ , and final component-wise LIM vector  $\lambda$ 
8:   Record  $\max(\lambda)$ ,  $\text{p95}(\lambda)$ , and the full vector  $\lambda$  [A2 convergence audit]
9:   if  $\max(\lambda) < \tau$  then
10:      $\text{status} \leftarrow \text{Strict-Accept}$ ,  $I_{\text{st}} \leftarrow \{1, \dots, m\}$ ,  $I_{\text{tail}} \leftarrow \emptyset$ 
11:   else if  $\text{p95}(\lambda) < \tau$  then
12:      $\text{status} \leftarrow \text{P95-Accept}$ ,  $I_{\text{st}} \leftarrow \{j : \lambda_j < \tau\}$ ,  $I_{\text{tail}} \leftarrow \{j : \lambda_j \geq \tau\}$  [A2 robust acceptance]
13:   else
14:     continue [A3 reduce component count and refit]
15:   end if
16:    $R \leftarrow WK$ ,  $D \leftarrow R^\dagger$ 
17:   return  $A_\ell = (\mu, K, W, R, D, S, I_{\text{st}}, I_{\text{tail}}, \text{status})$ 
18: end for
19: return Reject-Layer with diagnostics from all attempted component counts [A3 failed fit]
```

The returned artifact separates two roles. The **reading map** $R = WK$ projects normalized, centered activations into signed component scores, $S = X_c^\ell R^\top$. These scores indicate when a component is active. We use them for non-Gaussianity analysis, top-example retrieval, human annotation, activation-pattern visualization, and sparse probing. Since ICA scores are signed, annotation inspects both positive and negative sides when needed. The **writing map** $D = R^\dagger$ maps component coordinates back to the normalized fitting space. Its columns are the activation-space directions associated with ICA components. We use these write directions for comparison with SAE decoder directions and for intervention-style experiments, including targeted probe perturbation and steering-style edits.

3.1. Activation Corpus

We evaluate ICA on GPT-2 Small, Gemma 2 2B, and Qwen 3.5 2B Base. For each model, we fit ICA independently at the embedding layer and at each residual-stream layer. The embedding layer provides a non-contextual view of token-identity structure, while residual-stream layers expose contextual representations formed during autoregressive processing.

We refer to each recorded residual-stream hidden-state vector as an activation. For residual-stream fitting, we collect one million post-block residual-stream activations from randomly sampled token positions in the Pile-10k training split for each model.³ These sites match the residual-stream locations used by the

³<https://huggingface.co/datasets/NeelNanda/pile-10k>

corresponding public SAE checkpoints. In implementation, we use the hidden states returned by the Hugging Face model interface.⁴

Documents are truncated to a context length of 1024, and token positions are sampled uniformly without replacement using a fixed random seed. We run the model on the truncated documents and record hidden states at the sampled positions, so each activation is computed with its proper left context rather than as an isolated token.

Let $x_i^{(\ell)} \in \mathbb{R}^d$ denote the residual-stream activation at sampled token position i and layer ℓ . For each layer, we form

$$X^{(\ell)} = \begin{bmatrix} x_1^{(\ell)}, \dots, x_n^{(\ell)} \end{bmatrix}^\top \in \mathbb{R}^{n \times d}, \quad n = 10^6.$$

This gives a task-agnostic activation corpus that is not tuned to any later case study or benchmark. For the embedding-layer analysis, we replace the contextual activation matrix with the model’s static input embedding matrix and pass it through the same fitting interface. The embedding matrices contain 50,257 rows for GPT-2, 256,000 rows for Gemma 2 2B, and 248,320 rows for Qwen 3.5 2B Base.

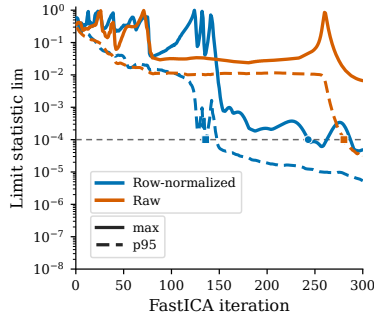


Figure 1: Layer-3 convergence diagnostics for GPT-2 Small. Blue curves use row-normalized activations and orange curves use raw activations. Solid curves report max-LIM, while dashed curves report p95-LIM. Full diagnostics are provided in Appendix A.

Rows	Max-LIM only		Max-LIM + p95 fallback	
	Raw	Norm.	Raw	Norm.
1k	2 / 3,431	9 / 2,177	2 / 3,431	9 / 2,177
3k	3 / 3,294	5 / 3,155	3 / 3,294	5 / 3,155
10k	2 / 3,499	6 / 3,206	4 / 3,383	6 / 3,206
100k	7 / 3,224	7 / 3,055	9 / 3,010	9 / 2,710
1M	2 / 3,492	8 / 3,107	6 / 2,954	10 / 2,741

Table 1: GPT-2 Small convergence yield. Each cell reports *accepted layers / total iterations* across 12 residual-stream layers. For the p95 fallback, we use max-LIM whenever it succeeds and p95-LIM only when the strict criterion fails. **Bold marks the best setting for each row.**

3.2. Row-Normalization

The first practical difficulty is the input geometry seen by the optimizer. When FastICA is applied directly to raw LLM activations, we often observe slow or unstable convergence. Prior work suggests two relevant intuitions. First, raw transformer activation spaces contain systematic outlier dimensions, rare massive activations, and attention-sink tokens, making raw activations highly anisotropic (Dettmers et al., 2022, Sun et al., 2024, Xiao et al., 2024, Han et al., 2025). Second, many activation-steering works treat directions in residual-stream space as meaningful objects for reading or controlling model behavior (Panickssery et al., 2024, You et al., 2026). We therefore normalize activations before fitting ICA, and surprisingly find that this recipe makes the fitting dynamics more stable and efficient.

Concretely, before the usual centering and whitening, each activation vector is normalized by its ℓ_2 norm: $r(x_i) = \frac{x_i}{\max(\|x_i\|_2, \epsilon)}$, where ϵ is a small numerical constant. ICA is then fit on the row-normalized matrix $\tilde{X}$,

⁴For the final layer of GPT-2 Small, the hidden state returned by Hugging Face is after the model’s final LayerNorm. We therefore use the raw block output before the final LayerNorm, preserving the same residual-stream convention used for earlier layers.

whose rows are $r(x_i)^\top$.

Figure 1 shows a representative convergence diagnostics example on GPT-2 Small layer 3. The normalized curves stay below the raw-activation curves through most of training. The normalized max-LIM curve reaches the 10^{-4} threshold within the iteration budget, while the raw max-LIM curve remains above it. Table 1 shows the same pattern at the layer level. Under the strict max-LIM criterion, normalization improves the accepted-layer count from 2 to 9 at 1k fitting rows, from 2 to 6 at 10k rows, and from 2 to 8 at 1M rows. It also usually reduces the total number of fitting iterations. The later annotation and intervention results show that the normalized components remain useful despite the loss of raw norm information.

3.3. Robust Convergence Acceptance

The second practical difficulty is convergence diagnosis. Standard FastICA uses a worst-case stopping rule:

$$\max_j \text{LIM}_j < \tau.$$

This rule accepts a fit only when every component has stabilized up to sign. We keep this strict criterion whenever it succeeds. However, in high-dimensional LLM activation fits, a small number of slow or oscillating components can dominate the maximum and cause the whole layer to be rejected. For exploratory interpretability, this is too conservative: many stable directions may be discarded because of a small unstable tail.

To make convergence less sensitive to the single hardest component, we record the full distribution of per-component LIM values. When the strict rule fails within the iteration budget, we use a softer fallback criterion:

$$p_{95}(\text{LIM}_j) \leq \tau.$$

This accepts a layer when almost all components have stabilized, while still flagging the remaining tail as difficult. For layers accepted by this fallback, we retain all components but attach a stability flag to each one. Components within the accepted 95% are marked as stable, while the remaining tail components are marked as unstable. We do not discard the unstable tail from qualitative inspection, since a slow-converging component may still exhibit a recognizable activation pattern. However, we preserve this flag in downstream audits and avoid using unstable components as primary evidence for convergence-sensitive claims.

Figure 1 illustrates this distinction. The dashed p95-LIM curves decrease more smoothly, while the solid max-LIM curves show sharper spikes and slower worst-case behavior. The aggregate effect appears in Table 1. With 1M fitting rows, raw activations accept only 2 of 12 GPT-2 layers under max-LIM alone, but 6 layers with the p95 fallback. With row normalization, the same comparison improves from 8 to 10 accepted layers and reduces total iterations from 3,107 to 2,741.

3.4. Adaptive Refit for Difficult Layers

Some layers still fail to meet the acceptance criterion within the iteration budget even after row normalization. We handle these cases with an adaptive refit procedure. We start with $m = d$ and halve m until the layer satisfies either the strict max-LIM criterion or the p95-LIM criterion, or until a minimum component count $m_{\min} = 16$ is reached. This gives each layer the highest accepted resolution under the same convergence standard.

Figure 2 summarizes the final accepted component counts across layers. Many layers converge at the full hidden dimension, while some layers require reduced component counts. The accepted component count is

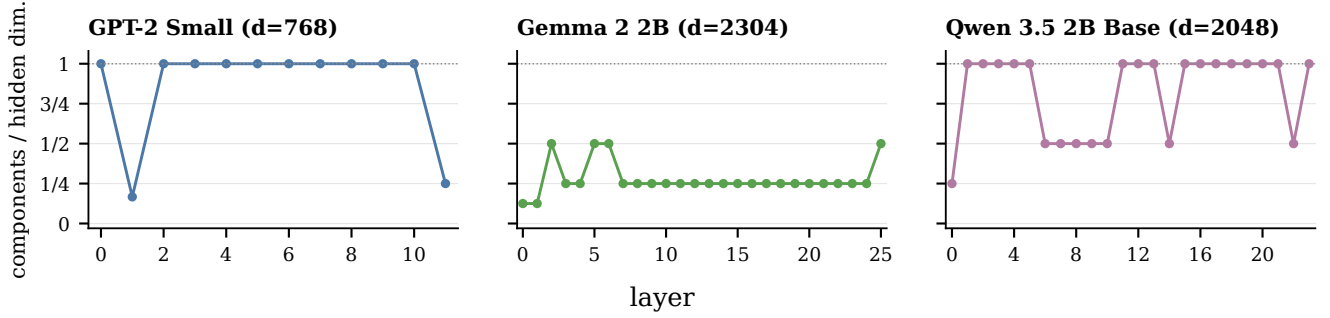


Figure 2: Final ICA component counts selected by the adaptive refit procedure. For each layer, we initially attempt to fit the full hidden dimension and reduce the target component count when convergence fails. Dashed horizontal lines indicate the hidden dimension of each model.

therefore also a diagnostic of fitting difficulty. A layer that accepts $m = d$ has a full-dimensional ICA basis. A layer that only accepts $m < d$ still provides a valid set of non-Gaussian directions.

4. What Does ICA Recover? Non-Gaussianity and Context Dependence

The previous section made ICA stable and efficient enough to fit on LLM activations. We now ask what kind of directions it recovers. A useful direction should pass two checks. First, it should be statistically exceptional rather than an arbitrary rotation of the activation space. Second, its exceptionality should correspond to recoverable structure in text, ranging from token-local patterns to broader contextual signals.

This section separates these two steps. We first verify that ICA components are substantially more non-Gaussian than random projections and SAE decoder directions (Section 4.1). We then introduce **effective receptive field (ERF)**, a diagnostic for each component (Section 4.2). ERF measures how much context is sufficient to recover a component’s activation at a target token, helping us characterize the context dependence of ICA components across layers and models.

4.1. Non-Gaussianity Separates ICA Directions from Random Projections

For a direction v and a set of row-normalized activations $\{\tilde{h}_i\}_{i=1}^n$, we measure the excess kurtosis of its scalar projection

$$z_i(v) = \langle v, \tilde{h}_i \rangle$$

as

$$\kappa(v) = \frac{\frac{1}{n} \sum_{i=1}^n (z_i(v) - \bar{z}(v))^4}{\left(\frac{1}{n} \sum_{i=1}^n (z_i(v) - \bar{z}(v))^2\right)^2} - 3, \quad \bar{z}(v) = \frac{1}{n} \sum_{i=1}^n z_i(v).$$

A Gaussian projection has excess kurtosis close to zero. Larger values indicate that the direction has a heavier-tailed or otherwise more exceptional projection distribution.

We compare three families of directions: random unit directions as a null baseline, SAE decoder directions as learned sparse-dictionary directions, and ICA score directions recovered by our fitted ICA maps. All three direction families are evaluated as projection directions on the same row-normalized activation distribution. This comparison therefore tests whether a direction is statistically exceptional under a common input geometry. Figure 3 shows a consistent separation across all three model families. ICA directions are substantially more

non-Gaussian than random directions and SAE decoder directions across layers, confirming that our fitted ICA components successfully recover the statistical objective of the method on LLM activations.

Interestingly, SAEs are not trained to maximize kurtosis, but their decoder directions already show elevated non-Gaussianity, suggesting that non-Gaussianity is a common statistical signature of many learned feature directions. This gives us a useful perspective on why ICA can serve as an interpretability lens: ICA makes this inductive bias explicit by directly searching for directions with exceptional projection statistics, rather than learning it implicitly through sparse reconstruction as in SAEs. In Sections 5–7, we show that these non-Gaussian directions from ICA are also human-interpretable and behaviorally useful under downstream feature evaluations.

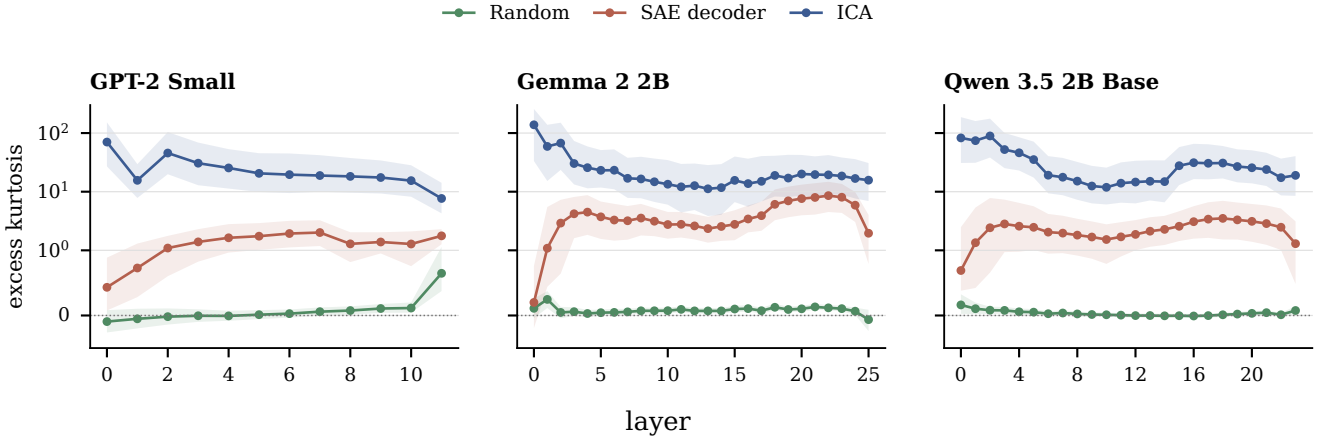


Figure 3: ICA recovers highly non-Gaussian directions in LLM activation space. For each model and layer, we project row-normalized residual-stream activations onto random unit directions, public SAE decoder directions, and fitted ICA score directions, then compute the excess kurtosis of each projection distribution. Across GPT-2 Small, Gemma 2 2B, and Qwen 3.5 2B Base, random projections remain close to Gaussian, SAE directions show elevated non-Gaussianity, and ICA directions are the most non-Gaussian, validating the statistical objective of the ICA fit. Statistics are computed over one million token positions from Pile-10k; random baselines use 32,768 Gaussian unit directions per layer. Lines show medians across directions, and shaded bands show interquartile ranges.

4.2. From Non-Gaussianity to Context Dependence

Non-Gaussianity tells us that a direction is statistically unusual, but not what textual structure makes it unusual. A high-kurtosis component may correspond to a local token pattern, a short phrase, or a broader contextual signal. We therefore need a way to ask whether ICA is only finding local detectors or also recovering components whose activation depends on context. To separate these cases, we introduce **effective receptive field (ERF)**.

ERF asks a simple question: how much left context is sufficient to recover the same signed component response at a target token? Small-ERF components can be recovered from the token itself or a short suffix, while large-ERF components require a broader preceding context.

Computing ERF. ERF is computed from a set of evidence examples for each component. For a target token position t in a sequence $x_{1:T}$, let $h_t(x_{1:t})$ be the residual-stream activation computed with the full available context, and let $s_j(h_t)$ be the signed score of ICA component j . We first identify the highest-scoring example

of component j in absolute value and record its sign. We then collect all examples with the same sign whose absolute score exceeds half of this maximum absolute score, forming an evidence set $\mathcal{E}_j$. In practice, this typically yields around 20 evidence examples per component.

For each evidence example (x, t) , we construct suffixes ending at the target token with lengths

$$x_{t-k+1:t}, \quad k = 1, 2, \dots, K_{\max}.$$

Let $h_t^{(k)}$ denote the final-token activation obtained from the length- k suffix. We then recompute ICA scores at this token. A suffix of length k is said to recover component j if the component remains among the top 15 components ranked by absolute score and preserves its original sign. Formally, let

$$R_j(x, t, k) = \mathbb{1} \left[j \in \text{Top}_{15} \left(\{ |s_r(h_t^{(k)})| \}_r \right) \wedge \text{sign}(s_j(h_t^{(k)})) = \text{sign}(s_j(h_t(x_{1:t}))) \right].$$

We search suffix lengths in increasing order and stop at the first successful recovery. The sample-level ERF is the shortest suffix length that recovers the component,

$$\text{erf}_j(x, t) = \min\{k \in \{1, \dots, K_{\max}\} : R_j(x, t, k) = 1\}.$$

If no suffix of length at most $K_{\max}$ succeeds, we assign $\text{erf}_j(x, t) = K_{\max}$ and mark the example as unrecovered within the tested window. In our experiments, $K_{\max} = 11$. The component-level ERF is the mean sample-level ERF over evidence examples:

$$\text{ERF}(j) = \frac{1}{|\mathcal{E}_j|} \sum_{(x,t) \in \mathcal{E}_j} \text{erf}_j(x, t).$$

ERF reveals a local-to-contextual spectrum. Figure 4 summarizes component context dependence as a layerwise distribution, showing many insights. First, ICA components span a broad local-to-contextual spectrum. Many components are recovered from the target token or a short suffix, but a substantial fraction require longer context. Thus, ICA is not merely finding lexical detectors; it also exposes directions whose activation depends on the surrounding text.

Second, ERF provides a quantitative lens on layer specialization. It shows that component scope changes with depth through a gradual shift in mixture proportions, not through a sharp handoff between layer types. Early layers are dominated by token-local components but already contain context-dependent directions, while later layers contain more medium- and long-context components yet still retain many token-local directions. Thus, layers are better described by the mixture of component types they contain than by a single local-to-global label. The distribution also shows a non-monotonic pattern. The largest-ERF components, including those recovered only at the maximum tested length or not recovered within the 11-token window, are often most prevalent in middle layers. This makes middle layers a useful target for inspecting context-dependent ICA components.

ERF offers one explanation for the kurtosis–interpretability link. Prior SAE work found that feature kurtosis is positively correlated with automated interpretability scores (Huben et al., 2024). This suggests that heavy-tailed feature activations are often easier for existing auto-interpretation pipelines to explain, but the correlation itself does not explain why. We use ERF to probe this link. Figure 5 shows a consistent negative relationship between ERF and excess kurtosis across all three models. Components with larger excess kurtosis tend to have smaller ERFs. These components are often activated by narrow, recurring

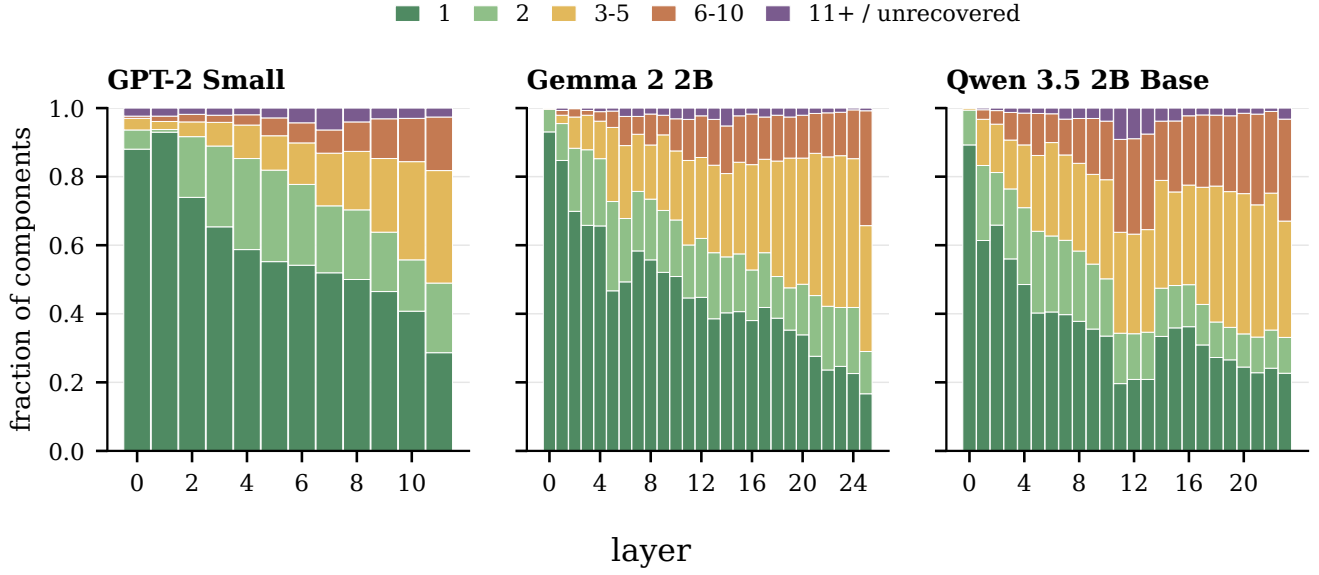


Figure 4: Layerwise distribution of effective receptive field (ERF). For each ICA component, we average the sample-level ERF over its evidence examples and bucket components by the resulting mean ERF. Each stacked bar shows the fraction of components in one layer. The final bin includes components not recovered within the 11-token window, reported as 11+

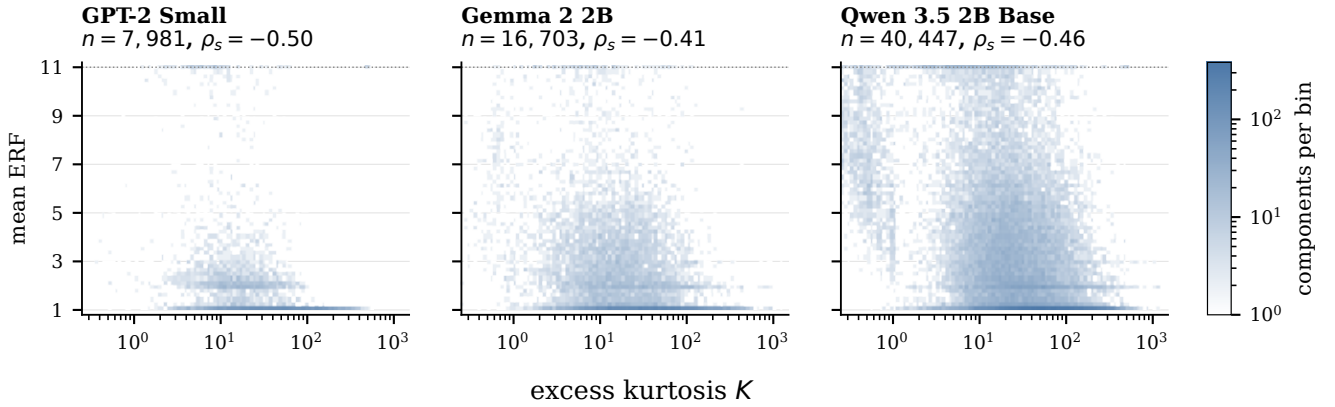


Figure 5: Relationship between effective receptive field (ERF) and excess kurtosis. Each cell counts ICA components with a given excess-kurtosis range and mean ERF. Across all three models, components with larger excess kurtosis tend to have smaller ERFs, with Spearman correlations ranging from -0.41 to -0.50 . This indicates that high-kurtosis components are typically more local, while broad-context components tend to have lower kurtosis and larger ERFs.

patterns such as tokens, surface forms, or short phrases, so their top examples expose the triggering pattern directly and are easier to summarize for both human annotators and LLM-based automatic interpretation methods. By contrast, larger-ERF components depend on broader context. Their activation conditions may not be visible at the target token or in a short local window, making them less sharply kurtotic and harder to explain from standard top-example views. Thus, ERF provides one operational account of why kurtosis can be informative for interpretation—high kurtosis is associated with locality, and locality makes top-example

explanations easier to form and verify. We treat this as a correlational diagnostic.

This interpretation also makes ERF useful for organizing annotation. Small-ERF components can often be analyzed from top examples alone, while larger-ERF components typically require context ablations or longer evidence windows. ERF therefore serves as a proxy for annotation difficulty. We use this ERF-guided strategy in our manual analysis in Section 5.2, and it may also help guide future automatic annotation systems.

5. Human Inspection of ICA Components

After quantifying the statistical structure and context dependence of ICA components, we now ask a more direct interpretability question: can these components be inspected, labeled, and reused in a controlled way? This section describes the inspection interface and annotation protocol, audits randomly sampled components, and then uses the resulting labels to analyze contextual and lexical structure.⁵

5.1. An Interactive Lens for Component Inspection

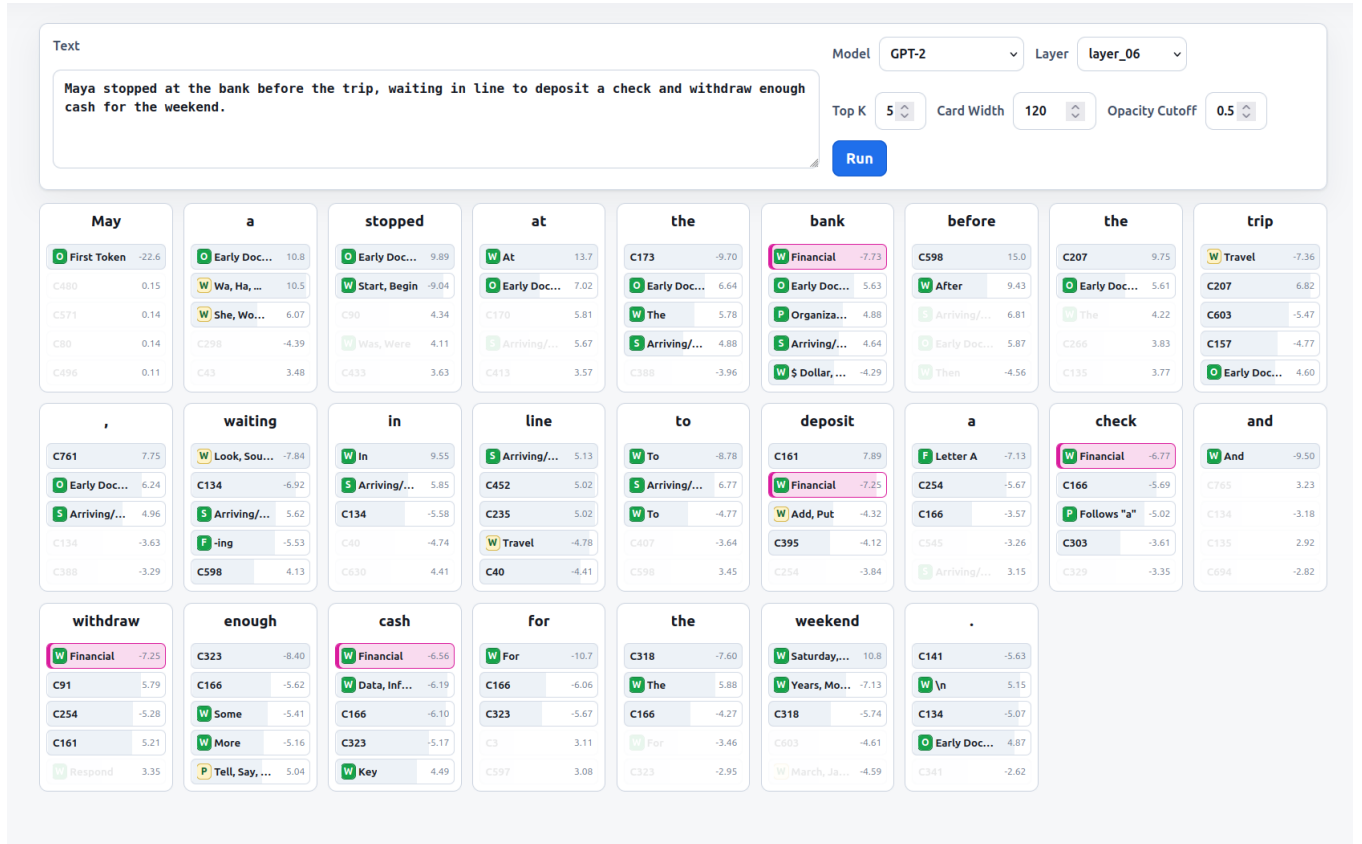


Figure 6: Interactive inspection view for GPT-2 Small layer 6. For each token, the explorer displays the strongest ICA components, signed scores, working labels, top examples, ERF, kurtosis, and annotation metadata used during component annotation. Additional screenshots for Gemma 2 2B and Qwen 3.5 2B Base are provided in Appendix F.

⁵We treat component labels as working hypotheses: short descriptions of the dominant pattern captured by a signed ICA direction, not ground-truth semantic definitions.

To support component interpretation, we built an interactive explorer that keeps the relevant evidence for each component visible during analysis. Given an input sequence, the interface displays the strongest ICA components at each token, ranked by absolute signed score. For each selected component, it shows the signed score, current working label, top-scoring examples, original document context, ERF, excess kurtosis, and annotation metadata. The explorer also links components across adjacent layers using cosine similarity between ICA directions. These links often provide useful clues when related patterns persist across depth.

The explorer is a quality-control tool for annotation. Its main role is to prevent labels from being assigned from a few isolated top examples without checking whether the component behaves consistently under additional contexts. The interface therefore supports both passive inspection of high-scoring examples and active hypothesis testing with targeted prompts.

5.2. Annotation Protocol

Given an activation h_t , we rank ICA components by absolute signed score $|s_j(h_t)|$. The magnitude indicates how salient component j is at the target token, while the sign selects one side of the ICA direction. Since ICA directions are identifiable only up to sign, annotations are sign-specific. The two sides of the same component may receive different labels when they correspond to different activation patterns.

Our annotation protocol is evidence-controlled and ERF-guided. For each component side, the expert annotator first checks the component’s ERF to estimate the expected scope of evidence. Low-ERF components can often be interpreted from the target token and its local context, while high-ERF components usually require broader document context or explicit context ablations. The annotator then inspects a fixed set of high-scoring examples, including the target token, signed component score, and surrounding text. When the pattern is ambiguous, the annotator also examines middle-range, near-zero, and opposite-sign examples to avoid explaining only the most extreme activations.

The annotation proceeds as an iterative hypothesis-testing process. The annotator first proposes a concise working label from the evidence examples. They then construct targeted prompts that should activate the component and matched counterexamples that should not. The explorer is used to check whether the component appears at the expected token positions with the expected sign. For context-dependent components, the annotator additionally performs context ablations by progressively removing preceding text and monitoring whether the component remains active. The label is revised until it is stable across top examples, targeted tests, and, when needed, context ablations.

Each labeled side receives a working label, a confidence level, and an interpretation type. Confidence is one of **high**, **medium**, **low**, or **unclear**. A high-confidence label consistently explains the observed behavior; a medium-confidence label captures a clear but harder-to-summarize pattern; a low-confidence label gives only a partial explanation; and an unclear label fails to provide a reliable interpretation. The interpretation type records the scale of the pattern: **form** components respond to surface features such as capitalization or initial letters, **word** components are tied mainly to token identity, **phrase** components respond to short local constructions, **sentence** components track sentence-level structure or topics, **long-range** components depend on earlier distant words or repetitions, **global** components reflect broader genre or discourse setting, **position** components respond mainly to token position, and **sophisticated** components appear to implement more conditional patterns.

5.3. Random Component Audit

To assess whether ICA components are practically inspectable, we conduct a random component audit. For each model, we uniformly sample 50 converged ICA components without replacement from the full component inventory, including embedding-layer components. For each sampled component, the expert annotators label the dominant signed side—the sign of the strongest absolute-score example. Empty labels and placeholder labels such as “?” are counted as unclear.

Model	High	Med.	Low	Unc.	W	P	F	S	G/L/Pos/X
GPT-2 Small	44	4	1	1	23	8	7	5	6
Gemma 2 2B	43	3	2	2	15	11	3	7	13
Qwen 3.5 2B Base	40	1	4	5	16	14	1	8	6
Total	127	8	7	8	54	33	11	20	25

Table 2: Random component annotation summary. We sample 50 ICA components per model and annotate the dominant signed side of each component. High/Med./Low/Unc. report annotation confidence. Type columns count the main interpretation type: word (W), phrase (P), form (F), sentence (S), and the union of global, long-range context, position, and sophisticated components (G/L/Pos/X). The complete annotation table appears in Appendix C.

Table 2 shows that most sampled components admit usable working labels under this protocol. Across 150 randomly sampled components, 142 receive non-unclear labels and 127 receive high-confidence labels. The labels are not limited to token identity. They include local form patterns, word categories, phrase templates, sentence-level constructions, global topics, position effects, and long-range or conditional patterns. We interpret these results as evidence of practical inspectability, because a randomly sampled ICA component often supports a concise and testable human hypothesis.

Table 3 shows representative prompt-level cases from the initial audit and how a working label is tested in practice. For each component, label-consistent prompts are paired with matched label-inconsistent prompts. We report the signed component score and its rank among components at the target token. Table 4 gives a compact view of representative audited labels across models.

We further test whether the first-stage expert labels remain supported under an independent expert audit using a prompt-based validation protocol. For each of the 127 high-confidence labels, a second expert auditor inspected the component evidence and the initial working label. The auditor then constructed matched contrastive prompts: label-consistent prompts intended to activate the component and label-inconsistent controls intended not to activate it. After running the frozen model and ICA decomposition on these prompts, the auditor assigned a 0–10 support score and wrote a revised label when the original description could be made more precise.

Table 5 shows representative rows from the secondary expert audit across all three models. Each row includes the initial label, the audit score, and the revised label suggested after the second inspection. The full audit table is provided in Appendix D. This secondary audit serves as a consistency check. Among the 127 audited labels, 121 are supported and the remaining 6 are partially supported; none are rejected. Moreover, 112 of the 127 labels receive a score of at least 8. The audit also shows that many initial labels can be sharpened, for example from “Gaming language” to “video-game discourse” or from “Scientific research, Citation” to “scientific reference-title final period.”

Component	Expected	Prompt	Score	Rank
L0/C192: After high, Word, negative side, ERF 1.0, $\kappa = 298$				
L0/C192	activate	I went outside to play <u>after</u> I finished my homework.	-22.447	1
L0/C192	activate	The puppy ran <u>after</u> the ball.	-22.700	1
L0/C192	not activate	I went outside to play <u>when</u> I finished my homework.	-1.017	18
L0/C192	not activate	The puppy ran <u>through</u> the ball.	+1.503	21
L7/C17: Scientific research / citation high, Sentence, negative side, ERF 2.0, $\kappa = 33$				
L7/C17	activate	Smith, J., & Lee, K. (2021).	-9.480	1
L7/C17	activate	Vaswani, A., et al. (2017). Attention is all you need.	-16.003	1
L7/C17	not activate	Smith and Lee (2021) is a fictional patent citation used here for illustration purposes only.	-2.051	14
L7/C17	not activate	In 2017, Ashish Vaswani led the team that published Attention Is All You Need, introducing the Transformer model.	+0.114	639
L10/C368: Gaming language high, Global, positive side, ERF 3.2, $\kappa = 31$				
L10/C368	activate	Don't fight before dragon unless our jungler has Smite up; save your ult for <u>their</u> engage.	+21.492	1
L10/C368	activate	If the tank does not kite during the second phase, the healers get overwhelmed <u>by</u> the damage.	+16.429	1
L10/C368	not activate	Don't leave before dinner unless everyone has arrived; save your announcement for <u>their</u> arrival.	+0.158	634
L10/C368	not activate	If the manager does not delegate during the busiest part of the project, the team gets overwhelmed <u>by</u> the workload.	-0.328	471
L8/C738: Repetition of a prior section header medium, Long-range, negative side, ERF 7.7, $\kappa = 23$				
L8/C738	activate	### Section 1: Discovery ... ### Section <u>2</u> : Analysis	-11.160	1
L8/C738	activate	Cat: maomao. Dog: wowowow? <u>Cat</u> : mamoa.	-7.496	2
L8/C738	not activate	### Discovery ... ### <u>Analysis</u>	+0.352	502
L8/C738	not activate	At 3:30 p.m., the race began, and the score was <u>3</u> :2 after the first round.	-1.737	55

Table 3: Representative prompt-level cases from the random component audit. Label-consistent prompts usually produce a large signed score and high rank at the target token, while matched label-inconsistent prompts reduce the score or rank. These controlled cases complement top-example inspection and make the working labels easier to falsify.

Model	Component	ERF	Type	Working label
GPT-2 Small	L0/C192	1.0	Word	After
GPT-2 Small	L9/C445	5.3	Long-range	Refer to a spokesperson
Gemma 2 2B	L13/C30	8.6	Sophisticated	Refer back to one of two alternatives
Gemma 2 2B	L10/C261	9.2	Long-range	Repetition and number increases
Qwen 3.5 2B Base	L18/C1794	3.5	Global	Study protocol
Qwen 3.5 2B Base	L22/C511	4.9	Sentence	Either/or, whether/or construction

Table 4: Representative labels from the random component audit. The examples span token-local, sentence-level, global, long-range, and conditional patterns. Labels are working descriptions under the inspection protocol, not ground-truth semantic definitions.

5.4. Case Studies: Reading Component Mixtures

The random audit asks whether ICA components can be labeled in isolation. We now ask how these labels behave when placed back onto concrete inputs. The case studies below use three complementary views. The first fixes a polysemous token and follows how its representation decomposes across layers. The second fixes

Model	Component	Initial label	Score	Revised label
GPT-2 Small	L0/C192	After	10	after token
GPT-2 Small	L7/C17	Scientific research, Citation	9	scientific reference-title final period
GPT-2 Small	L10/C368	Gaming language	9	video-game discourse
GPT-2 Small	L9/C445	Refer to a spokesperson	8	spokesperson/spokesman name reference
Gemma 2 2B	L24/C355	Turkish language	10	Turkish-language token/subword
Gemma 2 2B	L13/C30	Refer back to one of two recently introduced alternatives	9	reference to one member of a recent pair
Gemma 2 2B	L10/C261	Repetition and number increases	8	repeated identifier with increasing numeric suffix
Gemma 2 2B	L20/C263	Cryptography, Computer Security	8	cryptography / security context
Qwen 3.5 2B Base	L18/C1794	Study protocol	10	clinical study protocol/approval language
Qwen 3.5 2B Base	L19/C824	Follow, violate rules	10	rule compliance/violation language
Qwen 3.5 2B Base	L22/C511	Either or, Whether or, One way or another	9	alternative/concessive construction
Qwen 3.5 2B Base	L11/C1211	Because of, As a result of	9	causal preposition phrase

Table 5: Representative rows from the secondary expert audit of high-confidence labels. The table reports the initial working label assigned in the first-stage expert annotation, the 0–10 audit score, and the revised label suggested by the second expert auditor. The revised labels usually preserve the original interpretation while making it more specific.

two labeled components and traces their scores across sentences. The third removes context entirely and studies how embedding-layer ICA decomposes familiar lexical relations.⁶

5.4.1. Contextual Decomposition of a Polysemous Word

Polysemous words provide a controlled setting for studying contextual decomposition because the surface token is fixed while the surrounding evidence changes. We use a short paragraph containing four occurrences of *bank*. Two refer to a financial institution, denoted F1 and F2, and two refer to a river edge, denoted R1 and R2.

Maya stopped at the bank [F1] before the trip, waiting in line to deposit a check and withdraw enough cash for the weekend. The teller asked about her travel plans, and Maya said she was driving north to visit the old village by the river. By sunset she was sitting on the grassy bank [R1] with her shoes beside her, watching the water curl around stones and reeds. It amused her that the same word had followed her all day: first a bank [F2] with counters, accounts, and vaults, then a bank [R2] of earth holding the river in place.

Figure 7 is a component-level decomposition of four occurrences of the same token. Each column corresponds to one *bank* token in the paragraph. Each row corresponds to a layer. Within each cell, we list the five ICA components with largest absolute score at that token position. The bar length shows relative score magnitude

⁶Readers can reproduce and vary these examples in our interactive demo: [ICA Explorer case-study demo](#).

within the cell. The text label gives our working interpretation of the signed component side, and the badge records annotation type and confidence. Equivalent bank walkthroughs for Gemma 2 2B and Qwen 3.5 2B Base are provided in Appendix B.

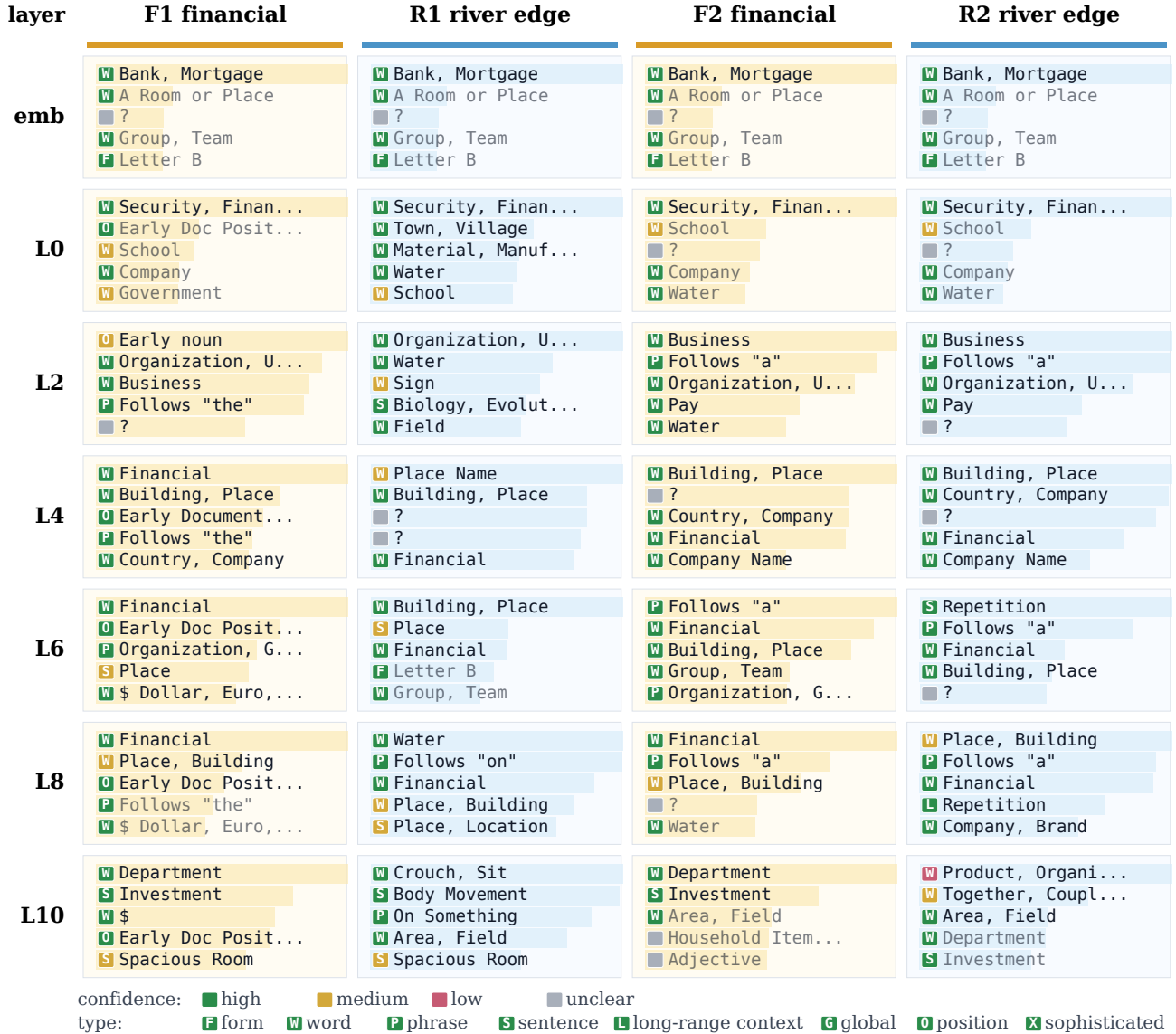


Figure 7: Contextual decomposition of a polysemous word in GPT-2 Small. We probe four occurrences of *bank* in one paragraph, including two financial-bank uses (F1, F2) and two river-edge uses (R1, R2). Each cell lists the five largest-absolute-score ICA components at the target token in a given layer. Bar lengths show relative score magnitude within the cell, and the legend decodes annotation type and confidence. Across layers, the same surface token is represented by different mixtures of lexical, syntactic, semantic, positional, and longer-context components.

The embedding row provides a useful control. All four columns are identical because the input embedding depends only on the token identity *bank*. The strongest embedding components include *Bank*, *Mortgage*, *A Room or Place*, *Group, Team*, and *Letter B*. Before contextual processing begins, all four occurrences share the

same lexical representation.

The residual-stream layers show how this shared lexical state separates under context. F1 occurs after the prefix “Maya stopped at the.” The later evidence, such as “deposit a check” and “withdraw enough cash,” is unavailable at the target token, yet the prefix already supports a financial reading. Later layers therefore contain components related to finance, organizations, companies, investment, and departments. This shows that the model can begin to disambiguate *bank* from left context alone.

R1 has a different mixture because the prefix already points to a physical scene. The paragraph has mentioned a village by the river, and the immediate prefix is “sitting on the grassy.” The active components shift toward labels such as *Water*, *Place Name*, *Building*, *Place*, *Crouch*, *Sit*, *Body Movement*, and *Area*, *Field*. These labels should be read as scene factors rather than a single river-bank feature. The token representation combines water, location, posture, movement, and outdoor space.

F2 returns to the financial sense, but its context differs from F1. The immediate prefix is “first a,” and the preceding paragraph has already introduced both senses of *bank*. The model therefore represents F2 as part of a repeated-word contrast while also activating financial components such as *Financial*, *Business*, *Organization*, *Pay*, *Department*, and *Investment*. The same cell also contains phrase-level components such as *Follows “a”*. This illustrates that one token representation can mix semantic content, local syntax, and discourse structure.

R2 is the easiest column to misread. A human reader identifies it as a river-edge use after seeing “bank of earth holding the river in place.” At the target token, however, the model has access only to the prefix “first a bank with counters, accounts, and vaults, then a.” This prefix explicitly mentions the financial sense and sets up a contrast. The finance-related components in R2, including *Financial*, *Department*, and *Investment*, are therefore expected under autoregressive conditioning. The cell reflects the model state at the target token, not the interpretation available after reading the following words.

This token-centered view gives two main lessons. First, ICA does not compress polysemy into one “financial versus river” direction. Each occurrence is represented by a changing mixture of lexical, phrase-level, semantic, positional, and longer-context components. Second, the mixture must be interpreted relative to the information available at the target position. The embedding row is token-driven, while later residual-stream layers incorporate the available prefix and discourse context.

5.4.2. Component Traces Across Sentences

The previous analysis fixes a token and asks how its representation decomposes across layers. We now take the complementary view. We fix two labeled components and ask how each component behaves across sentence positions. This is necessary because a component that appears in the top list for one token may not be a detector for that token alone. It may track a broader context that persists across several related tokens.

Sentence 1. She arrived at the library to study for her exam.

Sentence 2. Maya stopped at the bank before the trip, waiting in line to deposit a check and withdraw enough cash for the weekend.

Figure 8 shows this distinction directly. C67 is labeled as a financial component. In Sentence 1, which contains no financial context, its score stays close to zero. In Sentence 2, it becomes active around *bank* and remains active near *deposit*, *check*, *withdraw*, and *cash*. If C67 were only a detector for the word *bank*, we would expect a single spike at that token. Instead, the trace suggests a broader financial-context direction.

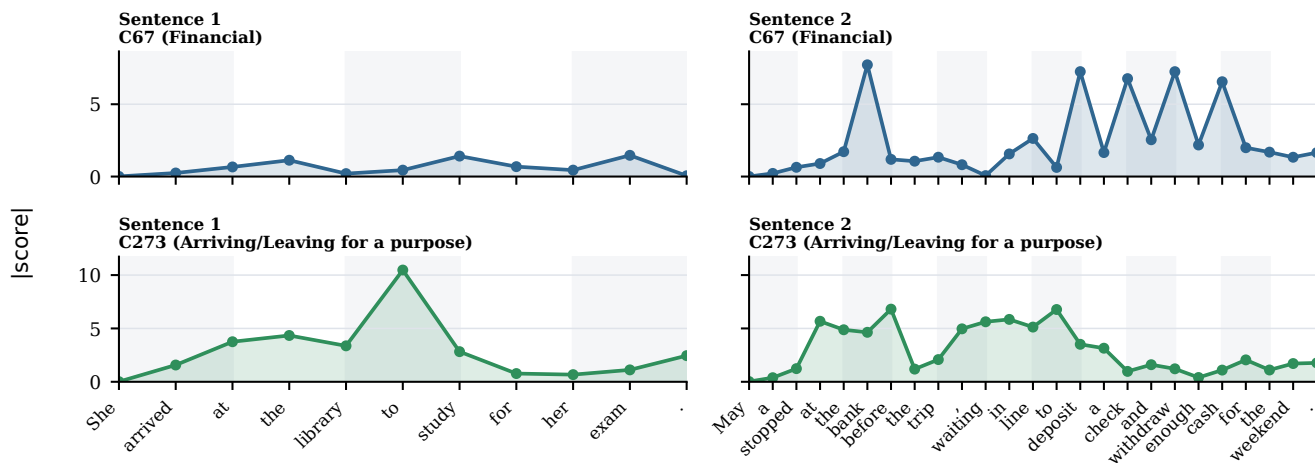


Figure 8: Sentence-level traces for selected GPT-2 Small ICA components. We plot raw absolute ICA scores across two sentences for layer-6 components C67 and C273. C67 tracks a financial-bank context and becomes active across several related tokens in Sentence 2. C273 tracks an arrival- or purpose-related construction and responds strongly to “arrived at the library to study” in Sentence 1, while also responding to related movement and purpose structure in Sentence 2. ICA directions can therefore trace context-sensitive signals across spans, rather than firing only at isolated tokens.

C273 illustrates the same point for a construction-level pattern. In Sentence 1, it rises around “arrived at the library to study,” which combines motion, destination, and purpose. The highest response occurs near the *to study* region, where the purpose relation becomes explicit. In Sentence 2, the same component also responds to related structure, including stopping at a place and waiting in line to perform an action. C273 is therefore better understood as an arrival- or purpose-related direction than as a detector for a single token such as *arrived* or *to*.

The insight from this trace view is simple. Some ICA components behave like local token or phrase detectors, but others trace contextual states across a span. Such traces help avoid overly narrow labels. A component whose largest score occurs near one token may still be responding to a larger construction. In our annotation workflow, we use these traces as hypotheses and then check them with targeted prompts and context ablations.

5.4.3. Embedding Components Recover Lexical Structure

The previous two case studies analyzed contextual representations. We now apply the same inspection protocol to token embeddings. This setting is a useful control because token embeddings and residual-stream activations live in the same d -dimensional representation space, but embeddings are fixed for each token. Embedding-layer ICA therefore asks which non-Gaussian directions organize lexical representations before any context enters the model.

Classical word-embedding analyses often summarize lexical relations through vector offsets, such as *king* – *man* + *woman* $\approx$ *queen*. This view is useful, but it compresses a relation into one displacement. ICA gives a more inspectable view. Instead of asking whether an analogy can be solved by one offset, we ask which readable directions are active in each token embedding.

Figure 9 is a component-level decomposition of static Qwen 3.5 2B token embeddings. Each box corresponds to one token, and each row inside the box is one high-scoring ICA component for that token embedding.

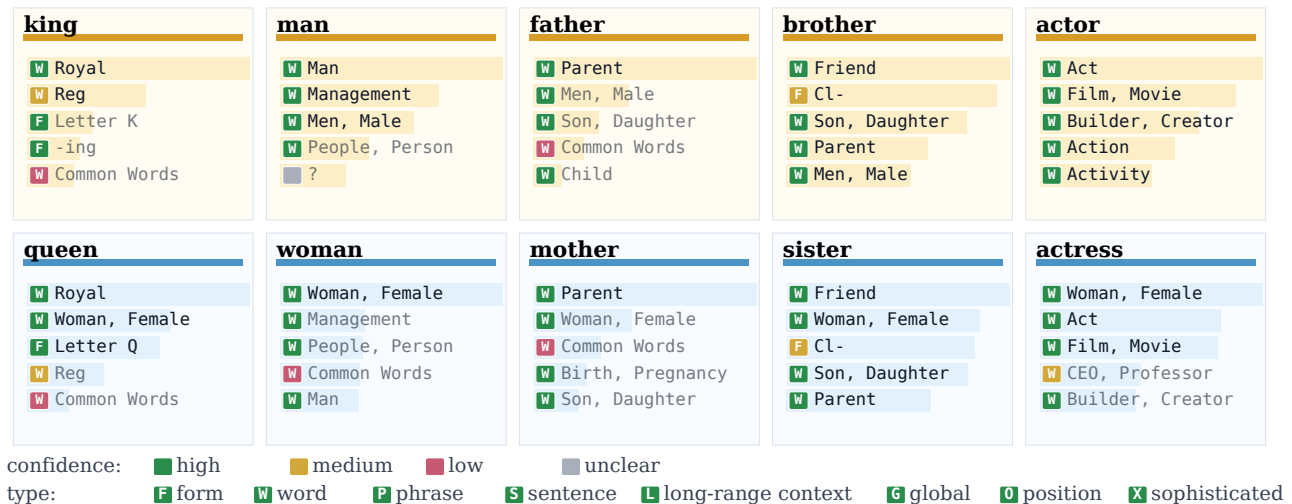


Figure 9: Embedding-layer ICA components for familiar analogy word sets in Qwen 3.5 2B Base. Each box corresponds to one token embedding and lists its five largest-absolute-score ICA components. Columns pair familiar masculine-coded tokens with their feminine-coded counterparts. Bar lengths show relative score magnitude within each box, and the legend decodes annotation type and confidence. The figure shows that familiar analogy pairs are supported by overlapping component directions, including royalty, gender-associated terms, family relations, professions, morphology, and common-token components.

Results for GPT-2 small and Gemma 2 2B are provided in Appendix B.

The first pattern is shared structure with extra factors. For *king* and *queen*, both tokens activate a *Royal* component, which recovers the expected royalty-related direction. Yet the two boxes also contain different additional components. *Queen* strongly activates *Woman, Female* and a surface-form component such as *Letter Q*, while *king* activates *Letter K*, morphology, and common-token components. Thus, the familiar pair is not represented only as “royalty plus gender.” It is represented as a mixture of semantic, gender-associated, morphological, and surface-form directions.

The second pattern is that different analogy families decompose in different ways. In the *man/woman* pair, both tokens activate person-related directions, while gender-associated components distinguish the pair. In the *father/mother* pair, both tokens share a *Parent* component and child-relation components. *Father* is associated with *Men, Male*, while *mother* is associated with *Woman, Female* and *Birth, Pregnancy*. The *brother/sister* pair shows a different organization. Both tokens activate directions such as *Friend*, *Son, Daughter*, and *Parent*, suggesting that sibling terms are represented through overlapping social and family-relation components rather than one isolated sibling axis.

The profession pair gives a third view of the same decomposition. *Actor* is dominated by components related to acting, film, creation, action, and activity. *Actress* shares several of these directions, including *Act*, *Film*, *Movie*, and *Builder, Creator*, but also prominently activates *Woman, Female*. This makes the relation more specific than a profession offset. The embedding contains both the shared occupation-related structure and the gender-associated component that distinguishes the pair.

This component view also makes asymmetries visible. *Queen* and *actress* prominently activate components labeled *Woman, Female*, while *king* and *actor* emphasize other directions among their top components.

Therefore, ICA can reveal asymmetric associations that are easy to hide when a relation is summarized by a single vector offset.

The main lesson is that embedding-layer ICA turns a coarse analogy view into a component-level explanation. Some directions are shared across a pair, such as royalty, parenthood, or occupation. Others distinguish the pair, such as gender-associated, morphological, profession-related, or common-token components. As in the residual-stream case studies, these labels are working descriptions rather than concept definitions. A component labeled *Royal* or *Woman*, *Female* is useful because it summarizes a recurring direction of variation in the inspected examples. This makes embedding-layer ICA a compact diagnostic for lexical structure before context enters the model.

6. Feature Utility under Sparse Probing and Intervention

The previous sections show that ICA directions are statistically exceptional and human-inspectable. We now ask whether they are useful as feature coordinates. A useful direction should concentrate task-relevant information in a small number of coordinates. Prior SAE Bench evaluations compare SAEs mainly against PCA as the classical linear baseline, but do not systematically evaluate ICA (Karvonen et al., 2025). We therefore evaluate ICA on two SAE Bench metrics: sparse probing for predictive feature utility and Targeted Probe Perturbation for selective feature intervention.

6.1. Sparse Probing

We first evaluate sparse probing, which tests whether feature activations contain concept-relevant information. For each concept dataset, SAE Bench ranks feature dimensions by how strongly their training-set activations separate positive from negative examples. It then trains a supervised linear probe using only the top- k ranked features.

As a concrete example, consider an AG News class such as *World*. SAE Bench treats examples from this class as positives and examples from the other classes as negatives. For each article, it runs the language model, collects residual-stream activations at the selected layer, masks special tokens, averages token-level feature activations into one vector for the whole text, and selects the k features whose training-set means best distinguish the positive class from the negatives. A linear probe is then trained on only these k features and evaluated on held-out articles.

For public SAE baselines, feature activations are obtained from the SAE encoder. For ICA and PCA, feature activations are the corresponding component scores. Since ICA and PCA scores are signed, we split each component into two nonnegative sides,

$$f_i^+ = \max(s_i, 0), \quad f_i^- = \max(-s_i, 0),$$

where s_i is the signed component score. The probe can then select the positive and negative sides separately.

We choose baselines to test four specific comparisons. **Public SAEs**, such as Gemma Scope and Qwen-Scope, are the standard high-capacity reference because they use large overcomplete dictionaries trained with reconstruction and sparsity objectives (Gao et al., 2025, Lieberum et al., 2024, Deng et al., 2026). **Matryoshka SAEs** test whether reducing the effective SAE dictionary size makes SAEs closer to ICA under compact feature budgets (Busmann et al., 2025). **ITDA** tests another training-light, non-SAE activation dictionary method (Leask et al., 2025). **PCA** tests whether non-Gaussianity is more useful than variance as a statistical signal for feature discovery (Abdi and Williams, 2010).

All methods are evaluated with the same sparse-probing datasets, feature budgets, and train/test splits. We average over two representative layers for each model: GPT-2 Small layers 6 and 10, Gemma 2 2B layers 12 and 20, and Qwen 3.5 2B Base layers 12 and 20. Since released Matryoshka SAE checkpoints are available for Gemma 2 2B, we compare them only on Gemma 2 2B layer 12.

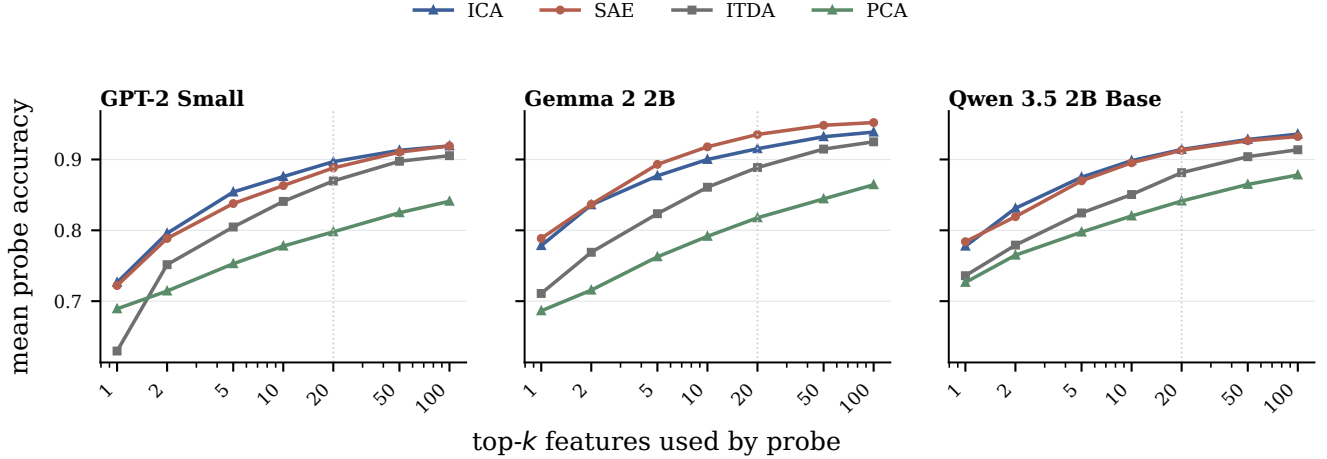


Figure 10: SAEBench sparse-probing performance for ICA, public SAE, ITDA, and PCA representations. For each representation, SAEBench ranks features by class contrast on the training set and trains supervised probes using only the top- k ranked feature activations. Each curve reports test accuracy averaged over the eight default sparse-probing datasets and two evaluated layers for that model. SAE features are produced by the SAE encoder, while ICA and PCA use signed component scores split into two nonnegative sides.

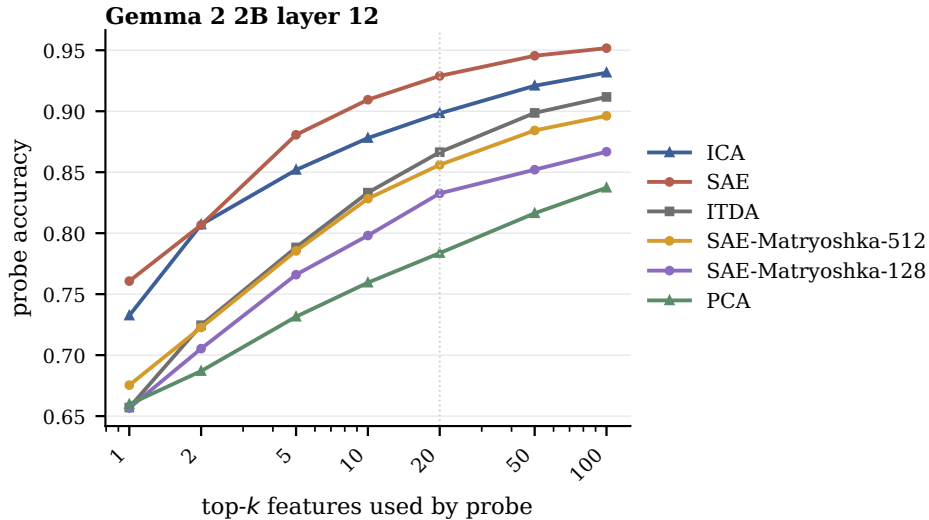


Figure 11: SAEBench sparse-probing comparison for Gemma 2 2B layer 12. We compare PCA, ICA, ITDA, prefix-restricted Matryoshka SAE variants, and the full Gemma Scope 16k SAE.

Figure 10 shows that ICA is a strong feature representation across all three model families. ICA remains competitive with public SAE dictionaries and consistently outperforms PCA and ITDA. Figure 11 further shows that compact Matryoshka variants with smaller dictionaries stay below ICA. These results show that: (1) ICA is competitive with public SAEs at concentrating concept-relevant information, (2) smaller SAE dictionaries do not close the gap by themselves, (3) ICA improves over the tested non-SAE alternative, and (4) non-Gaussianity is more useful than variance for sparse concept prediction.

6.2. Targeted Probe Perturbation

Sparse probing tests whether ICA coordinates are predictive. We next ask whether they can support selective interventions. We evaluate this with SAEBench Targeted Probe Perturbation (TPP) (Karvonen et al., 2025), an intervention-based feature-disentanglement metric.⁷ Since sparse probing shows that ICA and public SAEs are the strongest feature representations in our setting, we focus TPP on comparing ICA with public SAE dictionaries.

As a concrete example, consider the same AG News setting. SAEBench first trains one-vs-rest probes for all classes. For a target class such as *World*, it identifies the top- N features most associated with that class, zero-ablates those features, reconstructs the modified activation, and measures how all class probes change. A good feature representation should reduce the target-class probe score more than the non-target probe scores.

To evaluate ICA under the same TPP protocol as SAE features, we use the same two-sign ICA wrapper as in sparse probing. After intervention, the wrapper reconstructs the signed ICA scores as $s_i = f_i^+ - f_i^-$, maps the modified score vector back through the fitted ICA inverse, and returns the modified activation to the model.

Figure 12 shows that ICA is strongest relative to public SAEs at small-to-medium intervention budgets. Across all three model families, the intended effect of ICA rises quickly after ablating only a small number of features. At the same time, the unintended effect remains comparatively small. Therefore, when the goal is to inspect, edit, or intervene on a small set of task-relevant directions, a compact ICA basis can be advantageous. A small number of ICA components often captures enough class-relevant information to support selective interventions, without requiring the analyst to search through a large overcomplete dictionary.

At larger N , public SAEs become more competitive and can become stronger. This is expected, as SAE dictionaries are overcomplete and contain many more features, so increasing N gives SAEs more capacity to accumulate class-relevant latents. For ICA, the same N consumes a much larger fraction of the available compact basis, making broad disruption harder to avoid.

7. How ICA Relates to SAEs

Sparse autoencoders and ICA both expose directions in activation space, but they are designed around different objectives. SAEs learn large overcomplete dictionaries through reconstruction and sparsity losses. ICA instead returns a compact set of statistically non-Gaussian directions after normalization, centering, and whitening. Do they recover related structure, and if so, where do they differ?

7.1. Directional Overlap

For each ICA component, we find the public SAE decoder direction in the same model and layer with the largest absolute cosine similarity. This gives a simple measure of whether an ICA direction has a close counterpart in the SAE dictionary.

Figure 13 shows broad nearest-neighbor distributions across all three models. Many ICA components have a nontrivial SAE neighbor, so ICA is not unrelated to SAE. At the same time, many components are only

⁷We use TPP as a controlled, probe-based intervention diagnostic rather than as a standalone measure of feature quality. Recent work auditing SAEBench metrics cautions that TPP can be sensitive to benchmark settings and should not be over-interpreted in isolation (Chanin, 2026). We therefore interpret TPP together with sparse probing, human inspection, and ICA–SAE overlap analyses.

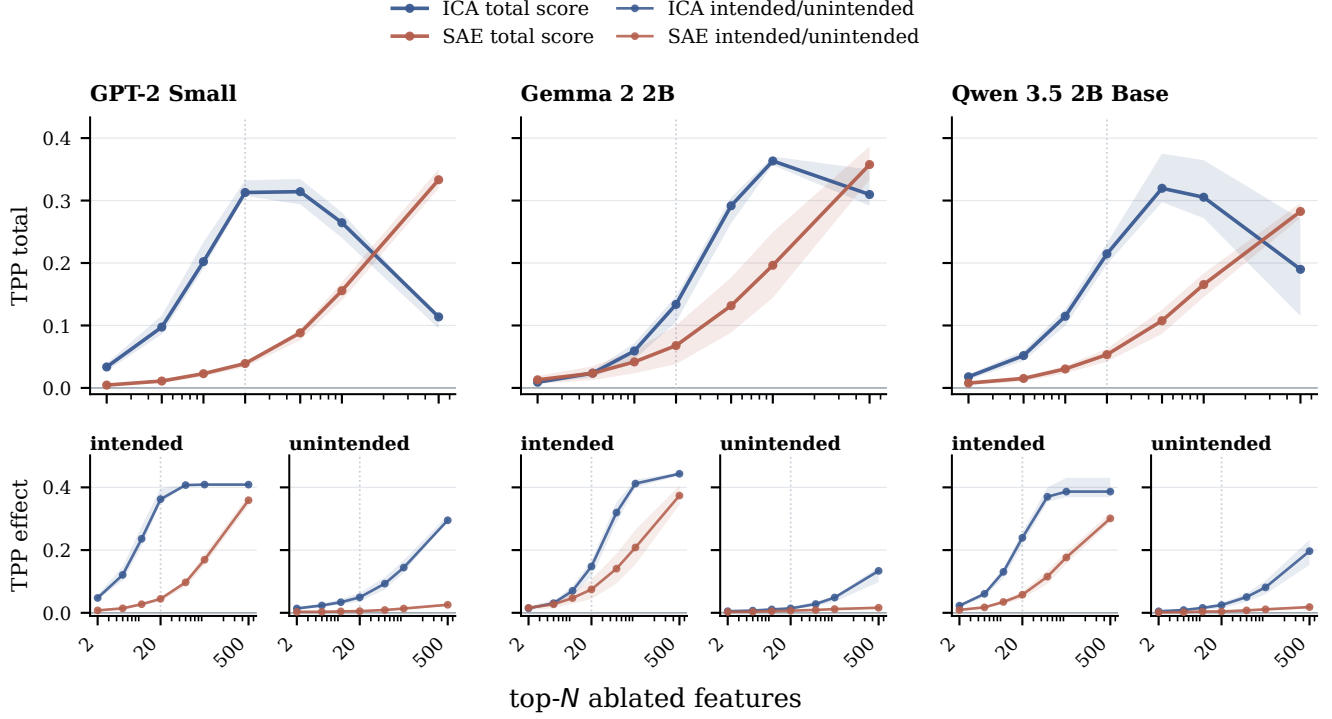


Figure 12: SAEBench Targeted Probe Perturbation (TPP) for GPT-2 Small, Gemma 2 2B, and Qwen 3.5 2B Base. The x-axis is the number of top-ranked features ablated for each target class. The top row reports the overall TPP score. The bottom row decomposes the same interventions into intended effects on the target-class probe and unintended effects on non-target probes. Scores are averaged over evaluated layers and the two TPP datasets, with shaded bands indicating interquartile ranges. ICA uses the two-sign SAE-compatible wrapper; SAE results use the public SAE dictionaries from the SAEBench comparison configs.

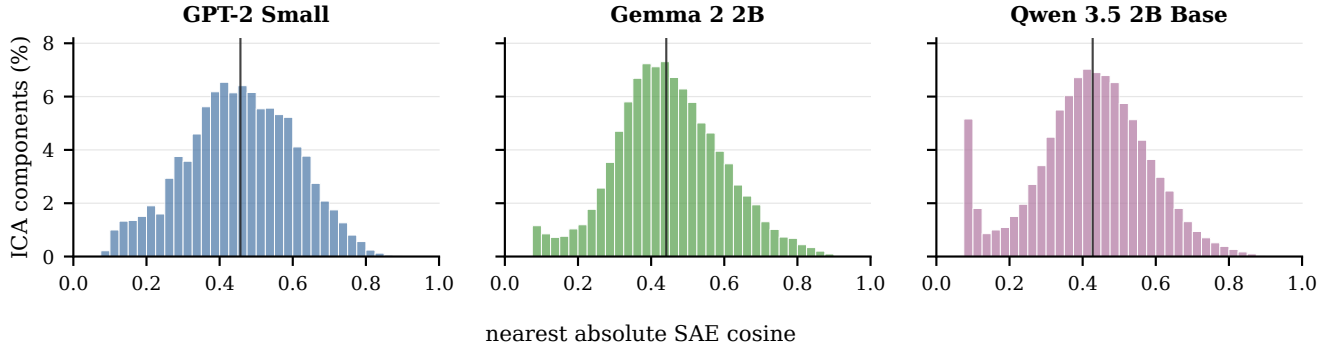


Figure 13: Nearest-SAE overlap for ICA components. For each ICA component, we report the maximum absolute cosine with any public SAE decoder direction in the same layer. Across all three models, most components lie in a moderate-overlap range, with both weakly matched and strongly matched tails. The distributions show partial agreement between ICA and SAE rather than a one-to-one correspondence.

weakly matched by any single SAE direction, so ICA is not merely a compact copy of public SAE dictionaries. Figure 14 shows that this pattern is stable across depth.

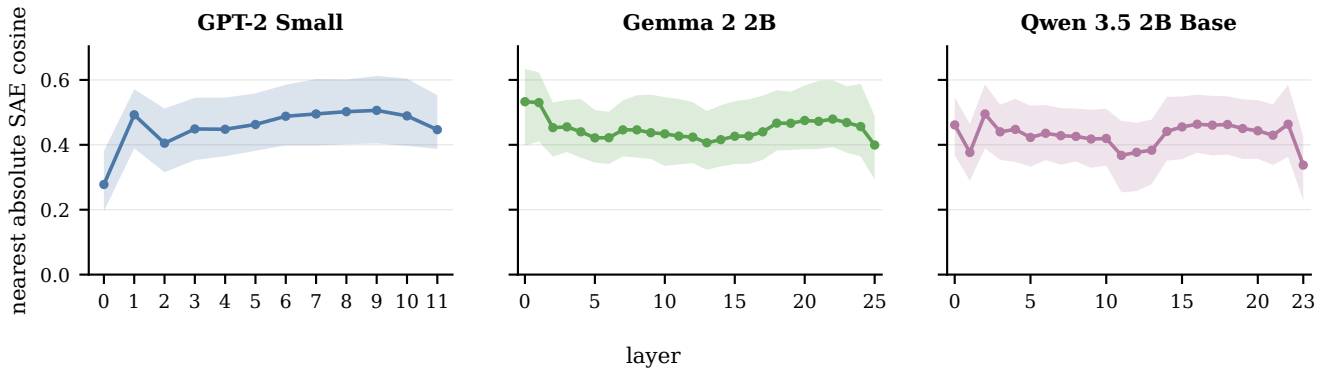


Figure 14: Layer-wise nearest-SAE overlap. Each point shows the median maximum absolute cosine between ICA components and public SAE decoder directions in one layer. Shaded bands show interquartile ranges. The median overlap remains moderate across depth, showing that partial ICA-SAE alignment is a consistent property of the decompositions rather than an artifact of a few layers.

We also check functional agreement. For each ICA component, we take its top-activation context and test whether the nearest SAE feature also activates on the same target-token activation. Table 6 shows concrete examples. The “SAE act.” and “SAE rank” columns measure whether the nearest SAE feature fires on the ICA component’s own top context. Low-overlap examples usually have inactive SAE neighbors, while mid- and high-overlap examples often activate their nearest SAE feature at the same target token. This suggests that high cosine is not only direction-level similarity; it often corresponds to similar activation behavior on real examples. These results provide evidence that high-overlap ICA components and SAE features can recover related interpretable structure.

A useful byproduct of this comparison is that it also lets us audit public SAE labels. We find that some public Neuronpedia labels are incomplete or overly narrow, and may not fully describe the feature’s behavior.⁸ For example, GPT-2 L10/C142 is labeled by our audit as conditional repetition, while its nearest SAE feature F29658 is labeled on Neuronpedia as numerical values and symbols in mathematical or programming contexts. However, the feature also fires on non-code contexts that match the ICA component’s repeated or template-like behavior, suggesting that the public label captures only part of the feature’s operational pattern. Similarly, Gemma L24/C510 is labeled by our audit as female-centered narrative, while Neuronpedia labels its nearest SAE feature F9501 as positive affirmations and encouragement. In controlled prompts, F9501 activates most strongly for “She’s passionate about books, travel, beauty, and all things cheese.” (99.5), less strongly for the male-substituted version (77.0), and still less for “He’s passionate about cars, football, and video games.” (40.75). This behavior is closer to a gendered narrative direction than to encouragement alone. From this perspective, ICA-SAE overlap can help cross-check and refine feature interpretations. It also highlights a limitation of current automated interpretability labels: many labels should be treated as hypotheses rather than ground truth.

⁸We refer to the public feature labels hosted on Neuronpedia, an online platform for browsing and annotating SAE features: <https://www.neuronpedia.org/>. Since these labels may be updated over time, we treat them as useful public hypotheses rather than fixed ground truth.

ICA	Label	Abs-top context	Nearest SAE	cos	SAE act.	SAE rank
GPT-2 low L0/C395	Water, Removed	...im and walk into the kitchen. I've already pulled a bottle of water	F10102	0.10	0	inactive
GPT-2 mid L0/C192	After	...f the tibial tray may remain exposed above the tibial plateau after	F10573	0.44	31.9	1
GPT-2 high L10/C142	Conditional repetition	string "You're pretty good.\$" Route110_Text_16EA2A:: @ 816EA2	F29658	0.75	46.1	14
Gemma low L6/C216	71 in No.	_) ASBCA N° 60315)) Under Contract No. HTC71	F9142	0.09	0	inactive
Gemma mid L12/C466	Accusation of bad behavior	...nger the lives of our law enforcement officers to promote a radical	F2168	0.43	9.87	33
Gemma high L24/C510	Female-centered narrative	...he referee, Carlos Ramos have wronged Serena Williams and took from	F9501	0.73	118	3
Qwen low L1/C834	Certain name	6, 1994. Decided July 8, 1994. Rear	F12249	0.08	0	inactive
Qwen mid L18/C1794	?	...ained from the patient included in the study, and the institutional	F3134	0.42	15.1	1
Qwen high L22/C511	?	...trigger, here's a post on how to determine whether they scrolled up	F8570	0.68	4.88	3

Table 6: Sampled ICA components and their nearest public SAE features. Rows are drawn from the preselected random-component audit set, choosing the lowest, median, and highest nearest-SAE cosine example for each model. The context column shows the ICA top-absolute-score example ending at the target token. SAE activation and rank measure whether the nearest SAE feature activates on that same target-token activation.

7.2. Activation Pattern Comparison

Direction overlap can reveal whether an ICA component has a nearby SAE feature, but it does not show whether their activations follow the same token-level pattern. To compare token-wise behavior, we use the same sentence for both decompositions. For GPT-2 Small, we select the top-1 ICA component by absolute score and the top-1 SAE feature by activation at each token. We then plot the union of the selected directions across the full sentence. Each row is normalized by its own maximum value in the sentence.

Figure 15 shows a clear difference in activation patterns. On the SAE side, selected features often peak at one token or a short span. This is consistent with the sparsity pressure in SAE training. On the ICA side, several directions vary more smoothly across neighboring tokens. In this example, ICA directions track the financial interpretation of *bank* across related words such as *deposit*, *check*, *withdraw*, and *cash*. This matches the behavior observed in our case studies, where ICA components often behave less like isolated event detectors and more like compact contextual factors. Full examples for Gemma 2 2B and Qwen 3.5 2B Base are provided in Appendix E.

8. Conclusion

We introduced ICALENS, a practical workflow for using ICA as a stable, efficient, and auditable lens on LLM representations. Rather than training another overcomplete dictionary, ICALENS normalizes activations, fits a compact GPU-parallel FastICA basis, and supports inspection, annotation, and evaluation of signed non-Gaussian directions. Across GPT-2 Small, Gemma 2 2B, and Qwen 3.5 2B Base, this workflow recovers human-interpretable components, links non-Gaussianity to context dependence through ERF, and yields useful feature coordinates under sparse probing and targeted probe perturbation.

Our results position ICA as a compact first lens for language-model activations, rather than as a replacement for sparse autoencoders. SAEs learn large overcomplete dictionaries optimized for sparse reconstruction, making them well suited for high-resolution feature discovery. ICALENS asks how much interpretable

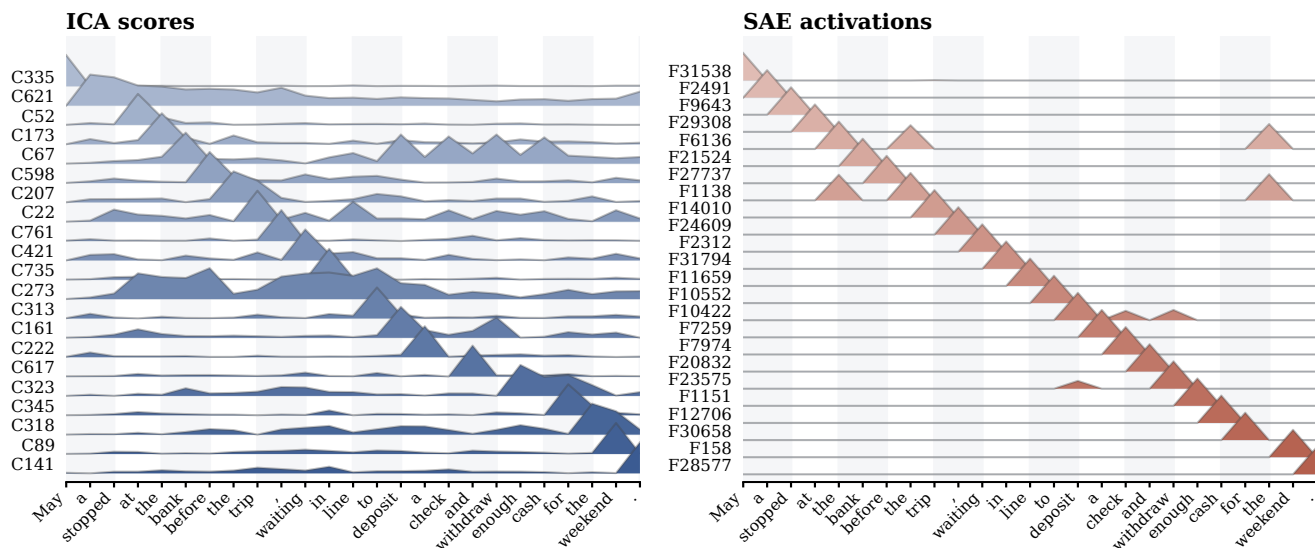


Figure 15: Token-wise ICA and SAE responses on the same GPT-2 Small sentence. At each token, we select the strongest ICA component by absolute score and the strongest SAE feature by activation, then plot the union of selected directions across the sentence. Each row is normalized by its maximum value to show within-direction response shape. SAE features mostly appear as localized activations, while ICA directions often vary across short spans, including the financial context around *bank*, *deposit*, *check*, *withdraw*, and *cash*. Sentence: “Maya stopped at the bank before the trip, waiting in line to deposit a check and withdraw enough cash for the weekend.”

structure is already visible from activation geometry, helping analysts decide where heavier dictionary learning is worth the cost.

9. Limitations and Future Directions

Compact ICA bases. The main algorithmic boundary is that standard FastICA is compact. It returns at most d components in a d -dimensional fitting space, so it cannot provide the large overcomplete vocabulary that modern SAEs are designed to learn. When a difficult layer is accepted with fewer than d components, the reading map remains useful for inspection and probing, while intervention-style edits rely on a pseudoinverse reconstruction. A natural next step is to explore higher-capacity ICA variants, including overcomplete ICA, preconditioned or adaptive FastICA, deflationary ICA, JADE, Infomax, extended Infomax, and heavy-tail-aware objectives. These methods may recover additional stable directions while keeping the objective simpler than full SAE training (Podosinnikova et al., 2019, Ablin et al., 2018, Miettinen et al., 2018, Cardoso and Souloumiac, 1993, Bell and Sejnowski, 1995, Lee et al., 1999, Spurek et al., 2018).

From states to transformations. Our experiments focus on residual-stream states, which reveal what information is present at a layer. A natural next step is to analyze what a layer changes. ICA could be fit to MLP outputs, attention outputs, residual updates, or shared bases across multiple layers. Such transformation-level analyses would provide a lightweight counterpart to model-wide SAE releases and related work on component-level tools for studying model computation (Lieberum et al., 2024, Deng et al., 2026).

Actionable analysis and annotation. Our main goal in this paper is to make ICA useful for human understanding: to help analysts quickly find, inspect, and test compact directions in model activations. Steering and automatic annotation are promising downstream directions. Because ICA is cheap to refit, it

may be especially useful for discovering task- or distribution-specific directions across different datasets, which is valuable for actionable mechanistic interpretability. Future steering applications should be evaluated against strong task-specific baselines (Wu et al., 2025, Kantamneni et al., 2025). Future automatic annotation systems could also build on the evidence used in our workflow—ERF, top examples, opposite-side examples, trace plots, prompt tests, and human audits—to produce more grounded component labels and turn automated descriptions into testable hypotheses (Paulo et al., 2024).

We will maintain an updated list of follow-up projects and open questions on the our website.⁹

⁹<https://liusida.github.io/ica-lens-paper/future-projects.html>

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A. Additional FastICA Fitting Diagnostics

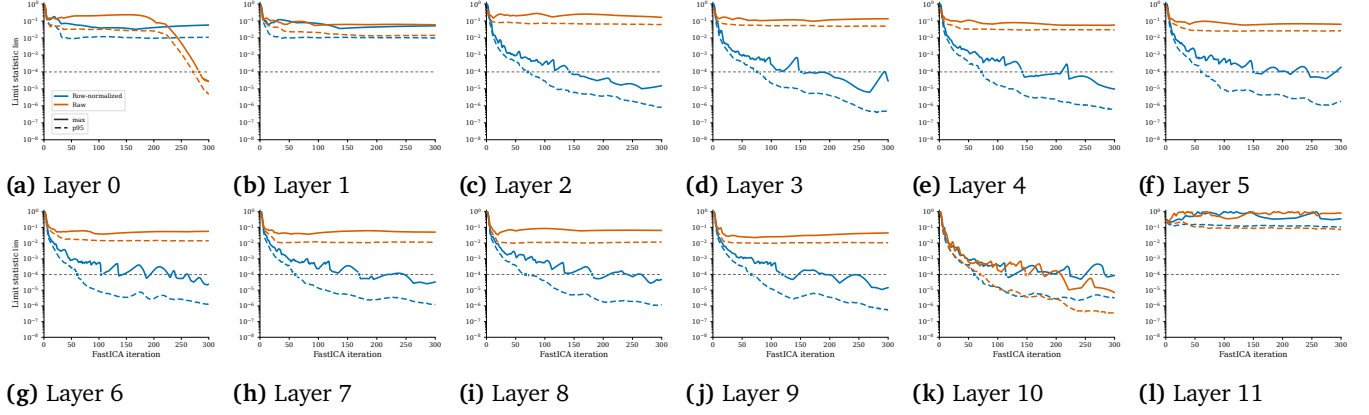


Figure 16: Layer-wise convergence diagnostics for FastICA on GPT-2 Small using 1k fitting rows.

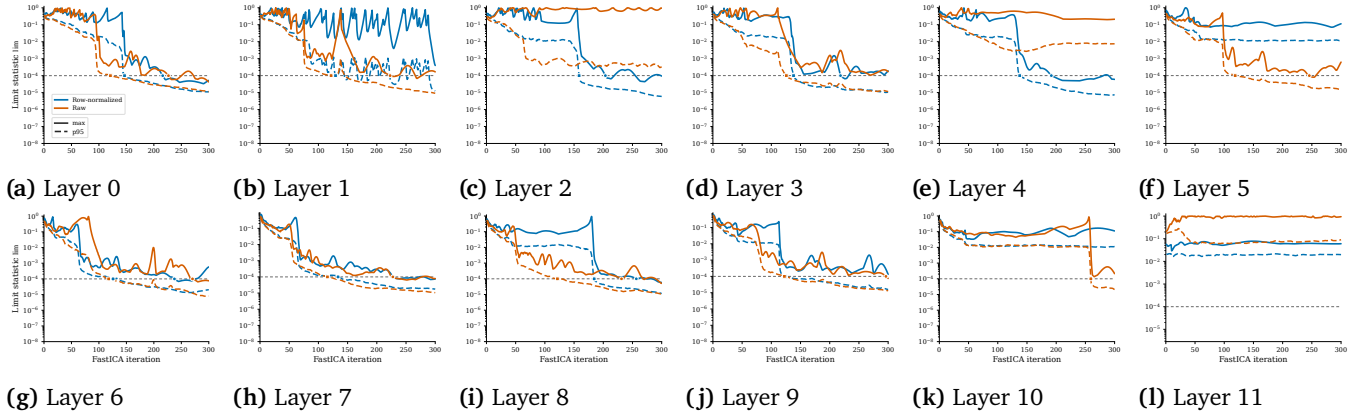


Figure 17: Layer-wise convergence diagnostics for FastICA on GPT-2 Small using 100k fitting rows.

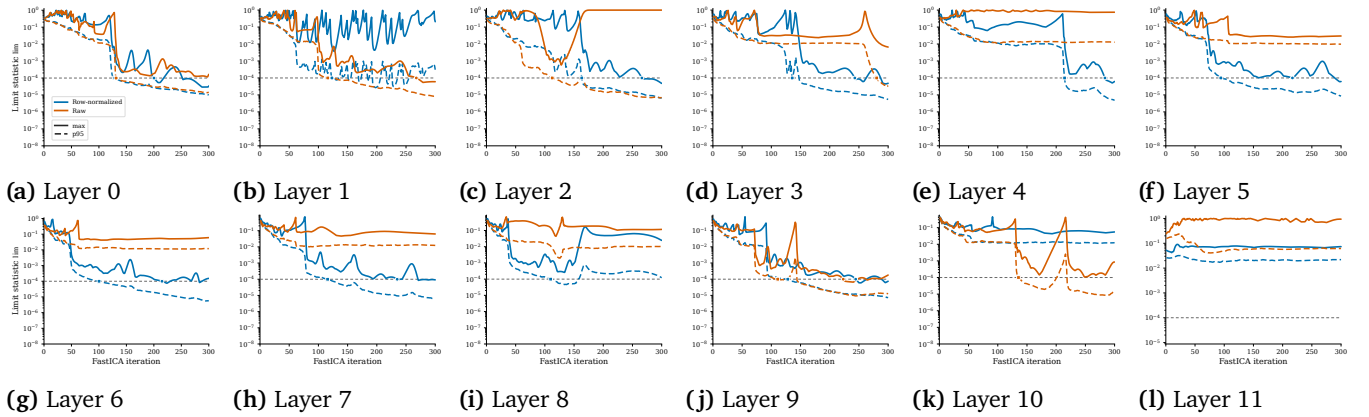


Figure 18: Layer-wise convergence diagnostics for FastICA on GPT-2 Small using 1M fitting rows.

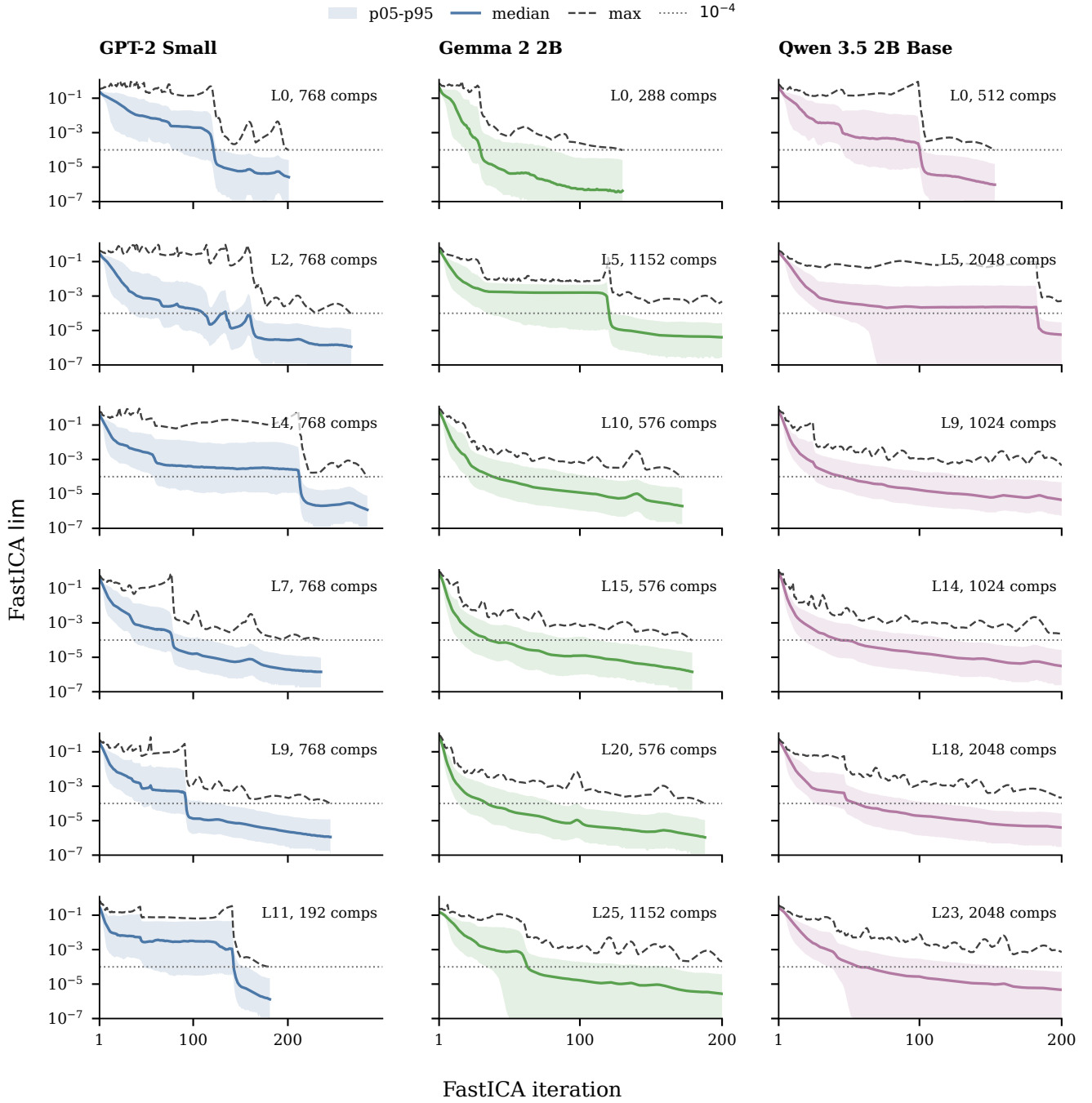


Figure 19: Representative FastICA convergence curves across model families. Each panel shows component-wise FastICA limit values over iterations for one fitted layer. Solid colored lines show medians, shaded regions show the 5th–95th percentile interval, dashed black lines show the maximum, and dotted horizontal lines mark the 10^{-4} convergence threshold.

B. Additional Human Interpretation Results

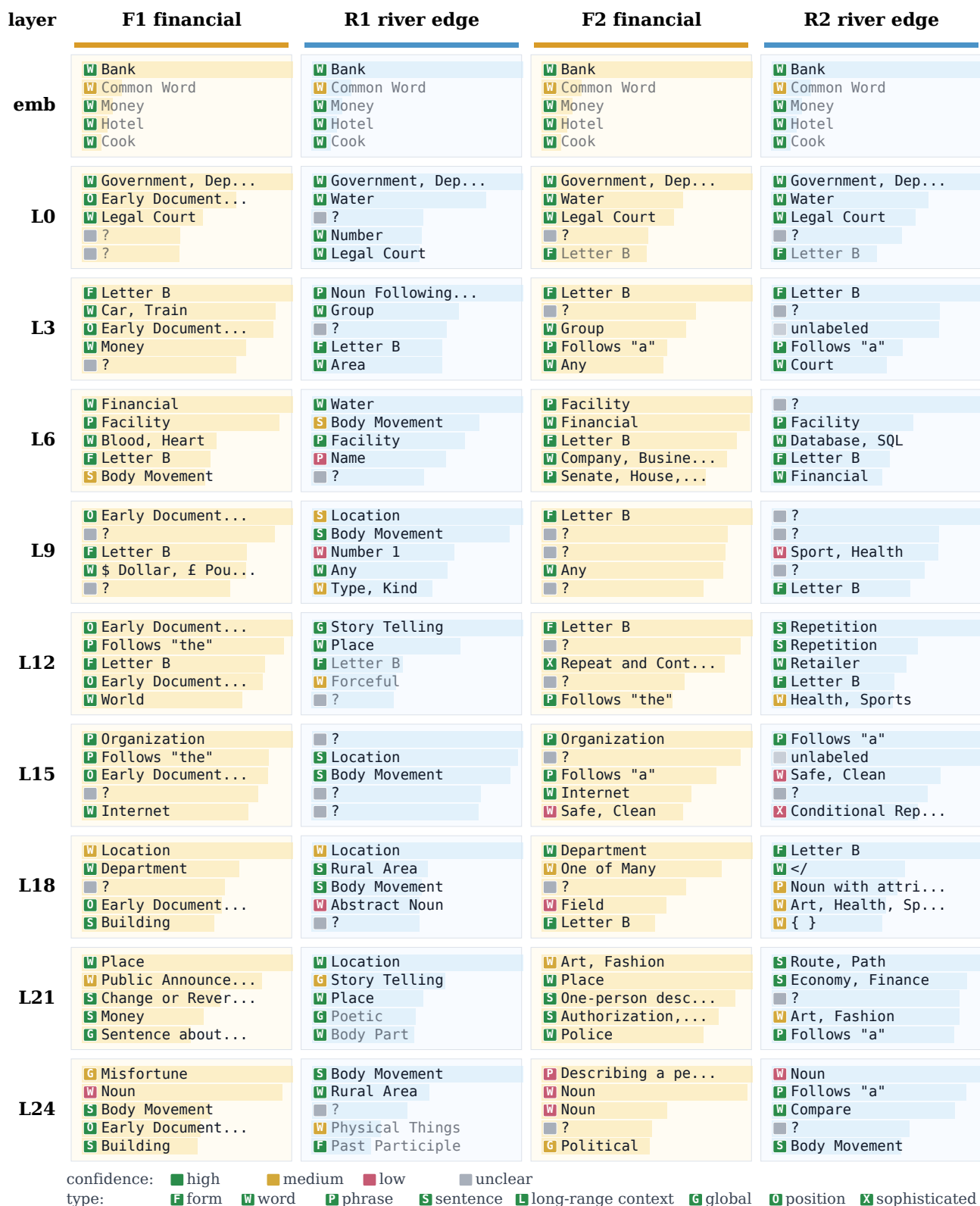


Figure 20: Contextual decomposition of a polysemous word in Gemma 2 2B.



Figure 21: Contextual decomposition of a polysemous word in Qwen 3.5 2B Base.

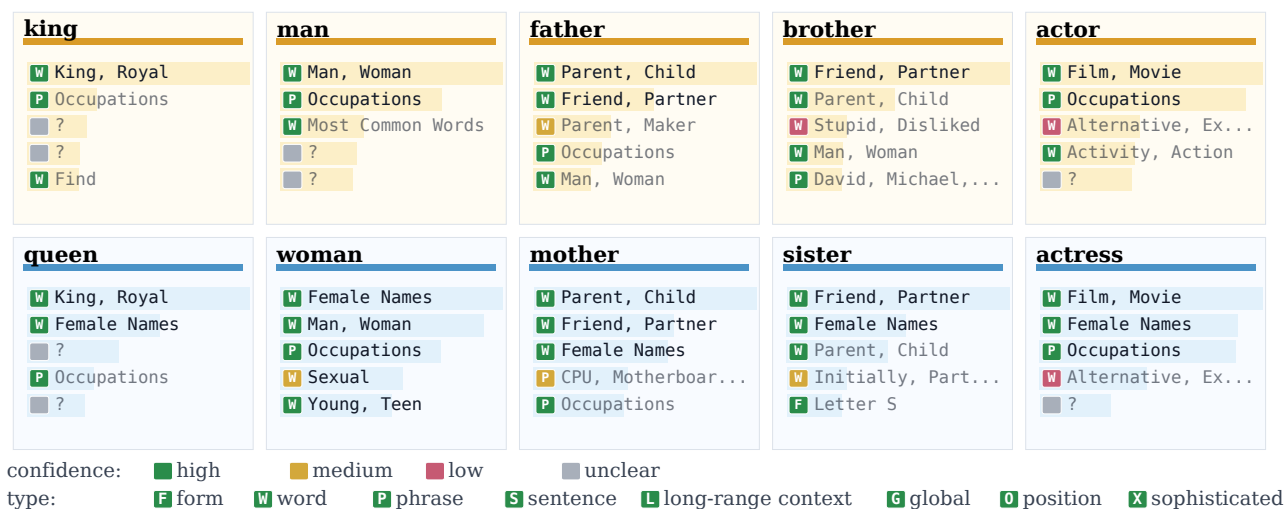


Figure 22: Embedding-layer ICA components for familiar analogy word sets in GPT-2 Small.

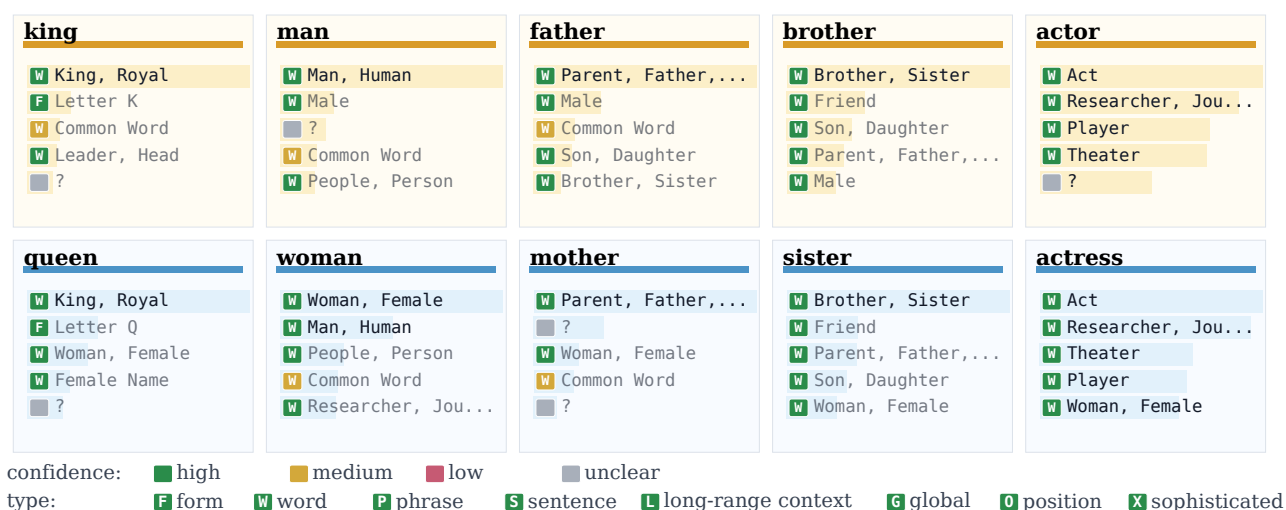


Figure 23: Embedding-layer ICA components for familiar analogy word sets in Gemma 2 2B.

C. Complete Random Annotation Tables

Table 7: Manual labels for the random ICA component audit. Rows are the selected components from `random_components_n50_seed0`. For each component, the table reports only the sign corresponding to its strongest absolute-score example, matching the annotation side most visible in the random-component audit. ERF is the mean effective receptive field measured by the short-context probe.

Layer/Component	Confidence	ERF	Type	Label	Excess kurtosis
GPT-2 Small					
L0/C6	high	1	Word	Arrest, Conviction	46
L0/C192	high	1	Word	After	298
L0/C267	high	1	Word	Any	194
L0/C366	high	1	Word	"	92
L0/C395	high	1	Word	Water, Removed	6.7
L1/C11	high	1	Word	Be, Been	31
L1/C97	high	1	Word	Also, Therefore	15
L1/C109	high	1	Word	\t, \r	23
L2/C302	high	1	Word	]	101
L2/C303	high	1	Form	Letter B	79
L2/C563	high	1	Word	En	80
L3/C257	high	1	Form	Letter Z	49
L3/C291	high	1	Word	Seem, Appear	29
L3/C330	high	1	Word	Very	129
L3/C413	high	1	Phrase	Inverted Is, Are, ...	64
L3/C449	high	1	Form	Letter U	50
L5/C15	high	1	Word	Social Media	26
L5/C696	high	1	Word	Building, Place	15
L6/C392	high	1	Word	Semicolon	92
L7/C140	high	1	Word	Dead, Killed	14
L7/C260	high	1	Word	With	81
L7/C492	high	1	Word	Look, Seem	33
L7/C766	high	1	Word	If	134
L8/C552	high	1	Word	Out, Down, Up	61
L9/C463	high	1	Position	Early Doc Position	11
L9/C621	high	1	Form	Mod-, Sub-	4.3
L10/C144	high	1	Form	Letter Z	24
L10/C375	high	1	Word	/,)/	15
L10/C576	high	1	Word	Try, Hard	9.6
L5/C11	high	1.4	Sentence	Sentence start	9.4
L9/C204	high	1.5	Phrase	Than, Comparison	30
L7/C254	high	1.5	Phrase	Thus, Therefore starting a clause	11
L4/C298	high	1.8	Phrase	Ongoing -ing	18
L6/C290	high	1.8	Global	Geopolitical conflicts	7.8
L4/C8	high	1.9	Form	D-	8.6
L2/C152	high	2	Phrase	Code whitespace	19
L7/C17	high	2	Sentence	Scientific research, Citation	33
L10/C74	high	2	Form	XML tag name	81
L5/C585	high	2.1	Phrase	i.e., Mean	12
L6/C164	high	2.1	Sentence	Citation	17
L8/C736	high	2.3	Phrase	Right, Permission, License to	3.0
L9/C118	high	3.1	Sentence	If-condition	7.9
L10/C368	high	3.2	Global	Gaming language	31
L9/C445	high	5.3	Long-Range Context	Refer to a spokesperson	24
L3/C294	medium	1	Word	Happy, Able	13
L6/C246	medium	2.2	Phrase	'in', 'the', or 'to' following 'tell' or 'say'	9.7
L10/C146	medium	5.4	Global	Natural area	11
L8/C738	medium	7.7	Long-Range Context	Repetition of a prior xxx: section header	23
L10/C142	low	8.3	Sentence	Conditional repetition	18
L2/C648	unclear	11+		?	-1.1
Gemma 2 2B					
emb/C1964	high	1	Word	kick	13
emb/C2034	high	1	Word	Smiley emoji	89

Continued on next page

Layer/Component	Confidence	ERF	Type	Label	Excess kurtosis
L0/C9	high	1	Word	Have	163
L1/C82	high	1	Word	"=" in code	124
L2/C302	high	1	Word	Does, Is, Will	134
L2/C341	high	1	Phrase	Two leading whitespaces	91
L2/C708	high	1	Word	Array	25
L2/C890	high	1	Word	Data	234
L5/C404	high	1	Word	Non-breaking space, U+00A0	25
L7/C351	high	1	Word	Do, Did	80
L16/C495	high	1	Word	Get	4.4
L19/C171	high	1	Form	Letter X	60
L20/C79	high	1	Word	Change	13
L20/C183	high	1	Word	Out	24
L5/C602	high	1	Word	Move	14
L9/C442	high	1	Word	Based, In response, Depend	7.1
L21/C399	high	1.1	Phrase	Adj. prefer after "be"	17
L7/C530	high	1.1	Phrase	Heart, Liver, Lung	27
L9/C79	high	1.1	Word	Director, President	21
L24/C355	high	1.2	Global	Turkish language	28
L21/C575	high	1.2	Phrase	Opening parenthesis in code	6.8
L15/C65	high	1.3	Phrase	Anatomical location	8.6
L10/C311	high	1.4	Phrase	Be the, Be a	23
L19/C210	high	1.6	Form	Letter V	12
L16/C527	high	1.8	Phrase	Family name	35
L18/C19	high	2.2	Sentence	Gas, Fluid	6.5
L6/C541	high	2.2	Phrase	What, How question	18
L16/C206	high	2.2	Global	Low-level programming language, C, ASM, ...	13
L6/C677	high	3.5	Sentence	If-condition	17
L9/C516	high	4.2	Sentence	Polynomial, Quadratic expression	2.6
L20/C263	high	4.4	Global	Cryptography, Computer Security	5.4
L22/C218	high	4.4	Sentence	Persistent deficiency despite expected improvement	5.9
L12/C466	high	4.6	Global	Accusation of bad behavior	4.3
L24/C372	high	5.2	Global	Immigration policy discussion	6.3
L18/C206	high	5.7	Long-Range Context	Same line ending in code	31
L8/C303	high	6	Sentence	At/In/During + time, something happened/changed	1.2
L19/C543	high	6.2	Sentence	Expecting additional parallel elements	29
L24/C510	high	6.6	Global	Female-centered narrative	10
L25/C854	high	7.5	Phrase	Citation marker	94
L23/C567	high	7.7	Global	UK civic info	6.8
L13/C30	high	8.6	Sophisticated	Refer back to one of two recently introduced alternatives	9.8
L10/C261	high	9.2	Long-Range Context	Repetition and number increases	35
L6/C775	high	11+	Long-Range Context	Repetition of prefix	19
L7/C106	medium	1.1	Sentence	Beginning of a new paragraph	7.7
L20/C101	medium	1.6	Phrase	Number > 10	40
L25/C92	medium	2.9	Global	Core issue in news	16
L6/C216	low	3	Phrase	71 in No.	0.5
L5/C453	low	3.5	Form	To-infinitive	0.4
emb/C59	unclear	1.7		?	0.2
L18/C33	unclear	11+	Long-Range Context	Apostrophe with unknown pattern	75
Qwen 3.5 2B Base					
emb/C930	high	1	Word	Where	107
emb/C984	high	1	Word	Sort	24
L0/C446	high	1	Word	Control	43
L1/C694	high	1	Word	Google, Big Tech Company	164
L1/C917	high	1	Word	Comma + a/the	152
L1/C1096	high	1	Word	Down	191
L1/C1604	high	1	Word	As (Capitalized)	186
L2/C511	high	1	Word	Shoot	138
L4/C1591	high	1	Word	Advertise	26
L7/C730	high	1	Position	Post-title article start	12
L7/C1009	high	1	Word	Without	35
L13/C1634	high	1	Word	Create, Develop	51
L16/C1572	high	1	Word	Only	150
L16/C1811	high	1	Form	Word fragment -e	24

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Layer/Component	Confidence	ERF	Type	Label	Excess kurtosis
L19/C808	high	1	Word	Chinese atmospheric literary narrative character	17
L21/C519	high	1	Word	>	49
L3/C508	high	1.1	Word	Register	14
L2/C1940	high	1.1	Phrase	End	268
L6/C647	high	1.3	Phrase	Last as in "last year/week"	44
L20/C189	high	1.3	Phrase	Total number, Amount	17
L13/C3	high	1.3	Phrase	Sensitive identity	14
L4/C1899	high	1.9	Word	Money, Fund	95
L13/C392	high	1.9	Sentence	Legal document format	56
L8/C522	high	2	Phrase	Is, Was, Has	32
L19/C824	high	2	Phrase	Follow, violate rules	23
L19/C999	high	2.4	Phrase	Math quantity, Math object	15
L11/C1211	high	2.4	Phrase	Because of, As a result of	13
L3/C77	high	2.5	Phrase	I feel I, You feel you, Feel like, ...	5.9
L13/C577	high	2.5	Sentence	"In" used in software context	10
L18/C280	high	3.1	Global	Reproductive health	20
L18/C1794	high	3.5	Global	Study protocol	46
L20/C970	high	4	Sentence	Motor, Electromechanical device	15
L18/C745	high	4.3	Global	Object in mechanical patent	21
L15/C258	high	4.4	Phrase	Age of a person or group	30
L22/C511	high	4.9	Sentence	Either or, Whether or, One way or another	5.0
L21/C145	high	4.9	Phrase	Citation (pandoc-style with dashes inside the ID)	29
L14/C707	high	5.3	Phrase	Legal case citation	16
L8/C189	high	5.3	Sentence	Number in elementary arithmetic	25
L9/C47	high	5.6	Sentence	Contrast, Opposition	12
L19/C526	high	8	Global	Javascript i18n code	17
L7/C671	medium	6.1	Phrase	Closing bracket in code	20
L1/C834	low	3.6	Phrase	Certain name	1.0
L13/C1679	low	4	Sentence	Context where medical condition or biological process happens	7.4
L6/C441	low	11+	Long-Range Context	Beginning of a repeated labeled/enumerated entry after intervening context	3.4
L23/C1250	low	11+	Sentence	Memorized string in code snippet	184
L7/C406	unclear	4.6		?	0.5
L23/C541	unclear	5.6		?	1.2
L5/C126	unclear	7.4		?	0.5
L3/C1037	unclear	7.5		?	0.5
L19/C646	unclear	10.3		?	11

D. Secondary Contrastive Label Audit

Table 8: Secondary audit summary for the high-confidence audited components from the random ICA component annotation study. Rows are drawn from `random_components_n50_seed0`. Score is the 0–10 secondary audit score.

Layer/Component	Initial label	Score	Suggested label
GPT-2 Small			
L0/C6	Arrest, Conviction	9	arrest / conviction / guilty legal context
L0/C192	After	10	after token
L0/C267	Any	10	any token
L0/C366	"	8	double quote token
L0/C395	Water, Removed	8	water token
L1/C11	Be, Been	9	be / been token
L1/C97	Also, Therefore	8	also / therefore discourse connective
L1/C109	\t, \r	10	tab / carriage-return token
L2/C302	]	10	closing square bracket token
L2/C303	Letter B	10	B / b token
L2/C563	En	8	en subword token
L3/C257	Letter Z	10	Z / z token
L3/C291	Seem, Appear	9	seem / appear token
L3/C330	Very	10	very token
L3/C413	Inverted Is, Are, ...	8	inverted/question auxiliary
L3/C449	Letter U	10	U / u token
L5/C15	Social Media	9	social-media platform/context
L5/C696	Building, Place	8	building / apartment place token
L6/C392	Semicolon	10	semicolon token
L7/C140	Dead, Killed	9	died / killed / dead token
L7/C260	With	10	with token
L7/C492	Look, Seem	9	look / seem / sound evidential verb
L7/C766	If	10	if token
L8/C552	Out, Down, Up	9	out / down / up particle
L9/C463	Early Doc Position	8	early-document opening subtoken
L9/C621	Mod-, Sub-	8	mod/sub prefix token
L10/C144	Letter Z	10	Z / z token
L10/C375	/,)/	8	slash token in URL/path/markup
L10/C576	Try, Hard	8	try / trying / tries token
L5/C11	Sentence start	9	sentence-initial capitalized token
L9/C204	Than, Comparison	10	than comparison token
L7/C254	Thus, Therefore starting a clause	9	thus / therefore consequence connective
L4/C298	Ongoing -ing	9	progressive/ongoing -ing form
L6/C290	Geopolitical conflicts	9	geopolitical conflict/diplomacy context
L4/C8	D-	9	D-initial subword/token
L2/C152	Code whitespace	7	code indentation/alignment whitespace
L7/C17	Scientific research, Citation	9	scientific reference-title final period
L10/C74	XML tag name	10	HTML/XML closing tag name
L5/C585	i.e., Mean	8	i.e. / that is to say / mean
L6/C164	Citation	8	reference-list title-final period
L8/C736	Right, Permission, License to	9	permission/right/license to
L9/C118	If-condition	8	then/comma after if-condition
L10/C368	Gaming language	9	Video-game discourse
L9/C445	Refer to a spokesperson	8	spokesperson/spokesman name reference
Gemma 2 2B			
emb/C1964	kick	10	Kick / kicks token
emb/C2034	Smiley emoji	7	Emoji / emoticon token
L0/C9	Have	8	have / has / had auxiliary
L1/C82	"=" in code	8	Quoted attribute assignment in markup
L2/C302	Does, Is, Will	10	WH-question auxiliary does/is/will
L2/C341	Two leading whitespaces	10	Two-space indentation token

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Layer/Component	Initial label	Score	Suggested label
L2/C708	Array	9	array token
L2/C890	Data	10	data token
L5/C404	Non-breaking space, U+00A0	10	Non-breaking space (U+00A0)
L7/C351	Do, Did	10	do / did auxiliary
L16/C495	Get	8	get / got / gotten
L19/C171	Letter X	10	X / x token
L20/C79	Change	9	change / changes / changed
L20/C183	Out	9	out token
L5/C602	Move	8	move / moved / moving
L9/C442	Based, In response, Depend	9	based / in response / depend construction
L21/C399	Adj. prefer after “be”	7	predicate adjective after be
L7/C530	Heart, Liver, Lung	7	cardiac / cardiovascular and organ terms
L9/C79	Director, President	6	institutional role title
L24/C355	Turkish language	10	Turkish-language token/subword
L21/C575	Opening parenthesis in code	8	opening call/index delimiter in code
L15/C65	Anatomical location	9	anatomical term/subword
L10/C311	Be the, Be a	9	article after be
L19/C210	Letter V	10	V / v token
L16/C527	Family name	10	surname after given name/title
L18/C19	Gas, Fluid	8	gas/fluid flow medium
L6/C541	What, How question	9	What/How question auxiliary
L16/C206	Low-level programming language, C, ASM, ...	8	low-level systems-code token
L6/C677	If-condition	9	If-clause predicate after if this/it/that
L9/C516	Polynomial, Quadratic expression	8	minus sign in polynomial/algebra expression
L20/C263	Cryptography, Computer Security	8	cryptography / security context
L22/C218	Persistent deficiency despite expected improvement	7	still/remain poor despite expectation
L12/C466	Accusation of bad behavior	8	accusatory / blame discourse
L24/C372	Immigration policy discussion	8	immigration-policy news discourse
L18/C206	Same line ending in code	7	repeated code/list line-ending pattern
L8/C303	At/In/During + time, something happened/changed	7	temporal transition after time expression
L19/C543	Expecting additional parallel elements	7	parallel/list element expectation
L24/C510	Female-centered narrative	8	female-character narrative context
L25/C854	Citation marker	6	pandoc-style citation marker digit
L23/C567	UK civic info	7	UK public-service/civic information prose
L13/C30	Refer back to one of two recently introduced alternatives	9	reference to one member of a recent pair
L10/C261	Repetition and number increases	8	repeated identifier with increasing numeric suffix
L6/C775	Repetition of prefix	8	repeated identifier prefix
Qwen 3.5 2B Base			
emb/C930	Where	10	where / Where token
emb/C984	Sort	10	sort / Sort token
L0/C446	Control	10	control token
L1/C694	Google, Big Tech Company	9	Google token / big-tech context
L1/C917	Comma + a/the	9	article after comma appositive
L1/C1096	Down	10	down token
L1/C1604	As (Capitalized)	10	capitalized As token
L2/C511	Shoot	10	shot / shoot token
L4/C1591	Advertise	9	advertising / ads token
L7/C730	Post-title article start	9	article-start token after scientific title
L7/C1009	Without	10	without token
L13/C1634	Create, Develop	9	create/develop verb
L16/C1572	Only	10	only token
L16/C1811	Word fragment -e	9	word-final e fragment
L19/C808	Chinese atmospheric literary narrative character	8	Chinese literary/poetic atmosphere character
L21/C519	>	8	literal > in markup/code
L3/C508	Register	10	register token

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Layer/Component	Initial label	Score	Suggested label
L2/C1940	End	10	end / End token
L6/C647	Last as in "last year/week"	9	last in temporal expression
L20/C189	Total number, Amount	9	aggregate count/amount expression
L13/C3	Sensitive identity	9	sensitive identity term
L4/C1899	Money, Fund	9	money / funds token
L13/C392	Legal document format	8	legal-document header formatting
L8/C522	Is, Was, Has	5	is/was after discourse setup
L19/C824	Follow, violate rules	10	rule compliance/violation language
L19/C999	Math quantity, Math object	8	scientific/mathematical quantity term
L11/C1211	Because of, As a result of	9	causal preposition phrase
L3/C77	I feel I, You feel you, Feel like, ...	8	pronoun after feel/felt like
L13/C577	"In" used in software context	8	in within code/software context
L18/C280	Reproductive health	7	reproductive/adolescent health context
L18/C1794	Study protocol	10	clinical study protocol/approval language
L20/C970	Motor, Electromechanical device	9	electromechanical motor/device context
L18/C745	Object in mechanical patent	9	mechanical patent object term
L15/C258	Age of a person or group	9	age number for person/group
L22/C511	Either or, Whether or, One way or another	9	alternative/concessive construction
L21/C145	Citation (pandoc-style with dashes inside the ID)	9	dash inside Pandoc citation/key
L14/C707	Legal case citation	7	legal case citation whitespace
L8/C189	Number in elementary arithmetic	9	digit answer/context in elementary arithmetic
L9/C47	Contrast, Opposition	8	contrast continuation token
L19/C526	Javascript i18n code	7	JavaScript/JSON locale object

E. Token-Wise Activation Patterns in Gemma and Qwen

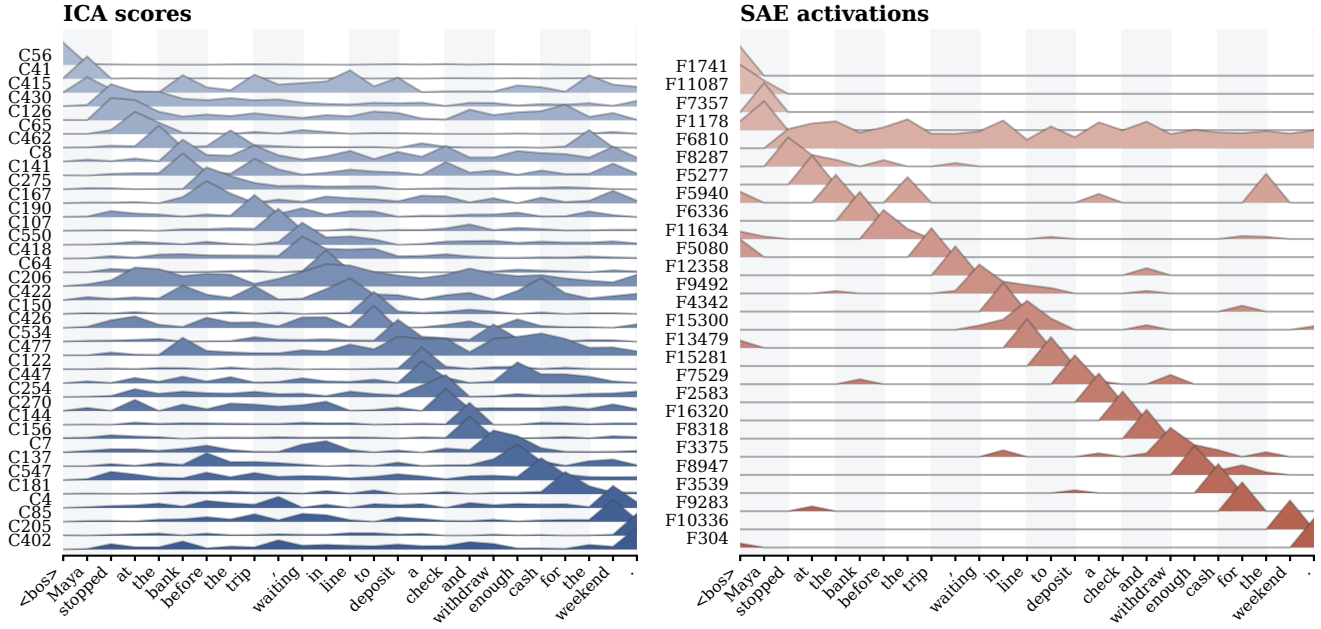


Figure 24: Token-wise ICA and SAE patterns for the same Gemma 2 2B sentence. For each token in the bank sentence, we select the top-2 ICA components by absolute score and the top-2 SAE features by activation, then plot the union of selected directions across all token positions.

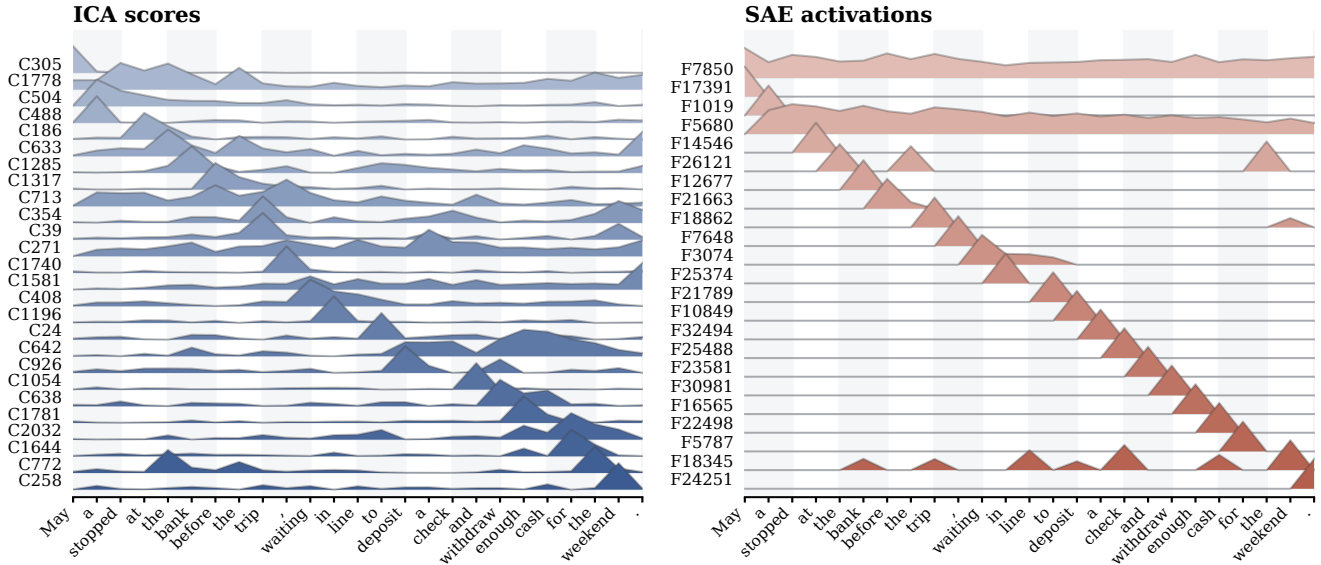


Figure 25: Token-wise ICA and SAE patterns for the same Qwen 3.5 2B Base sentence. For each token in the bank sentence, we select the top-2 ICA components by absolute score and the top-2 SAE features by activation, then plot the union of selected directions across all token positions.

F. Explorer Interface Screenshots

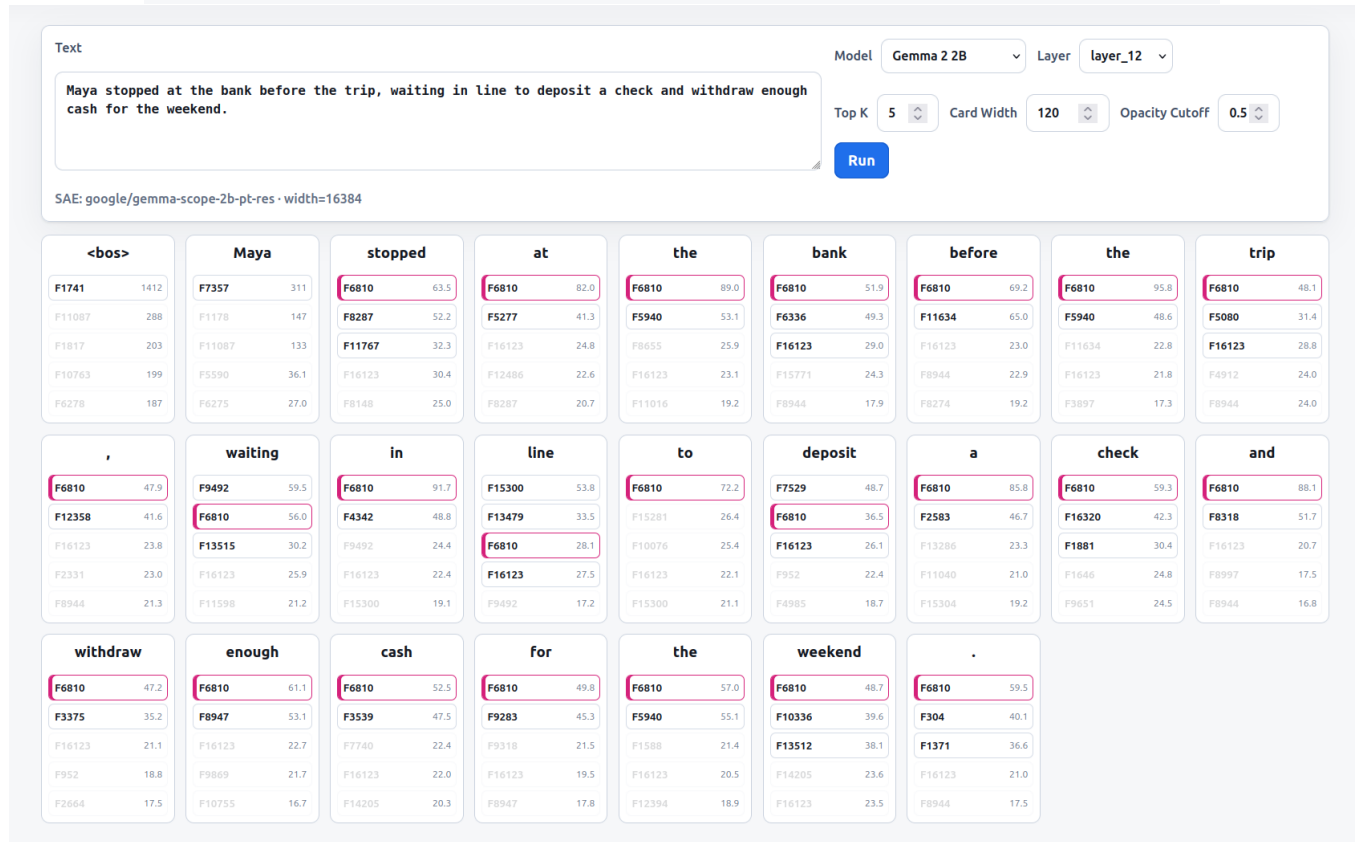
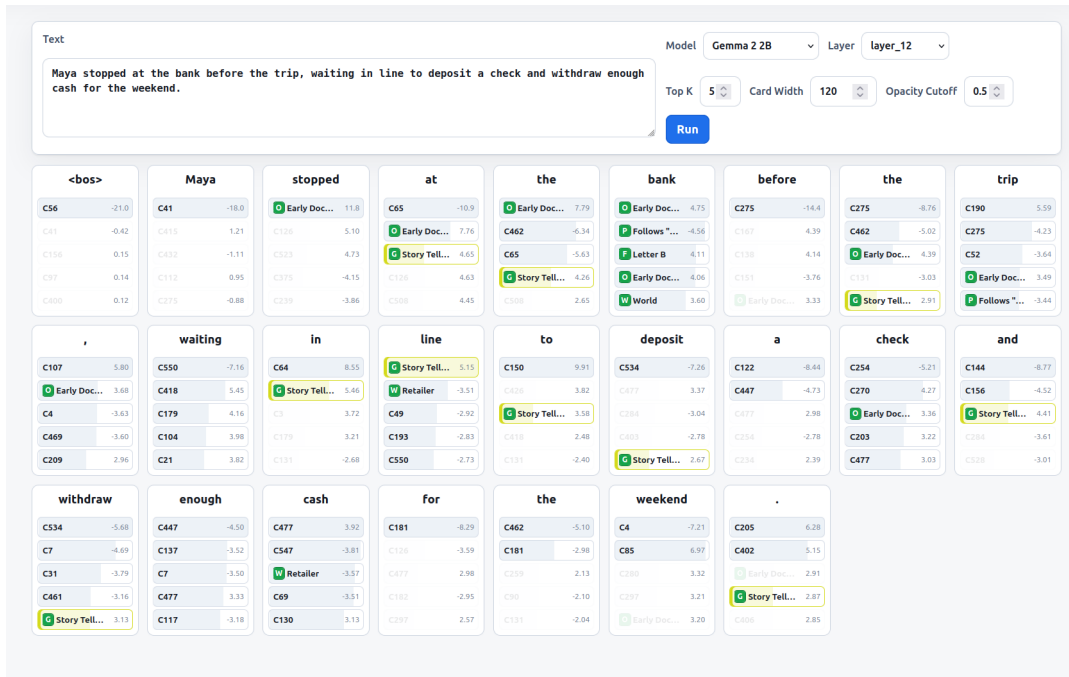


Figure 26: Explorer interface screenshots for Gemma 2 2B layer 12. Two screenshots are ICA Explorer and SAE Explorer.

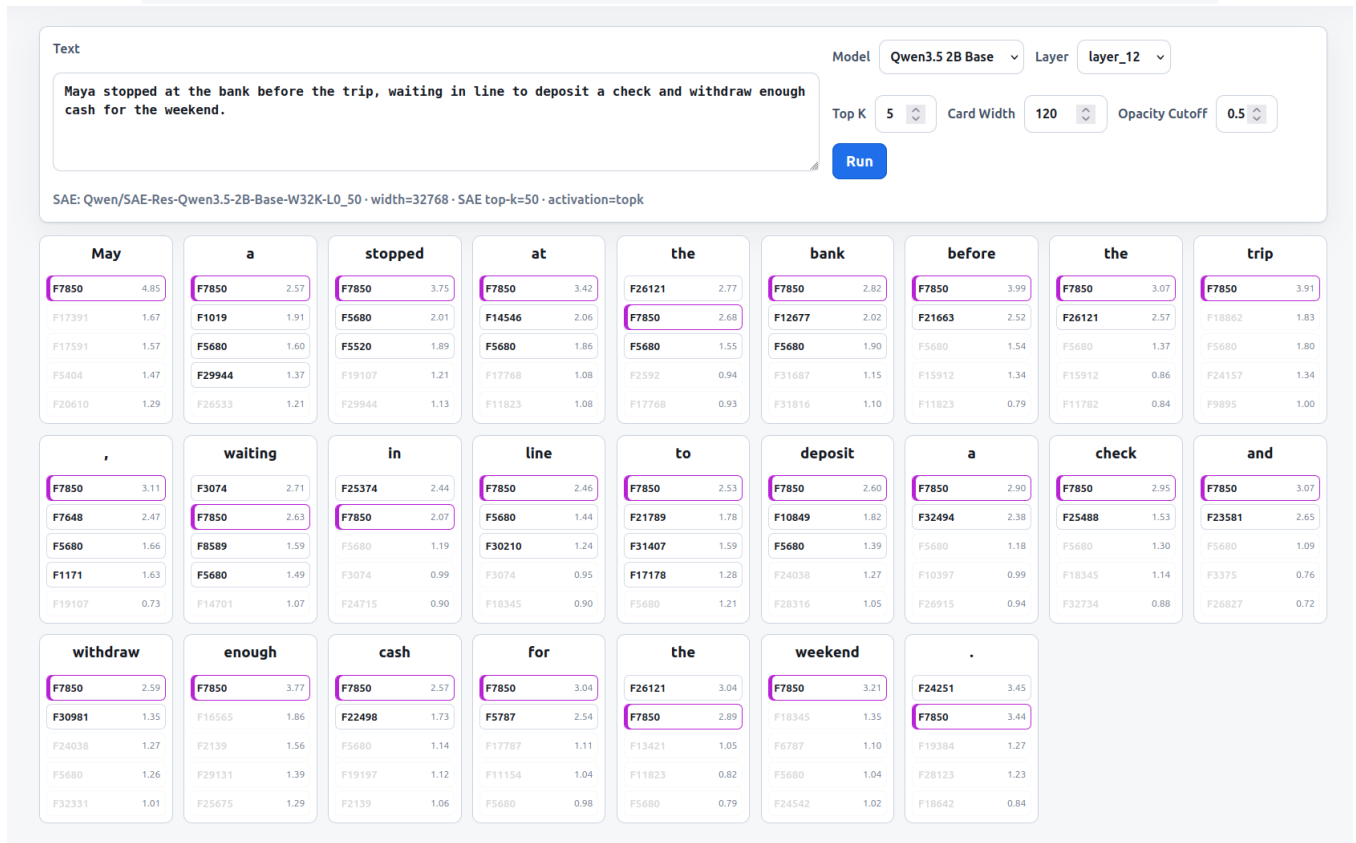
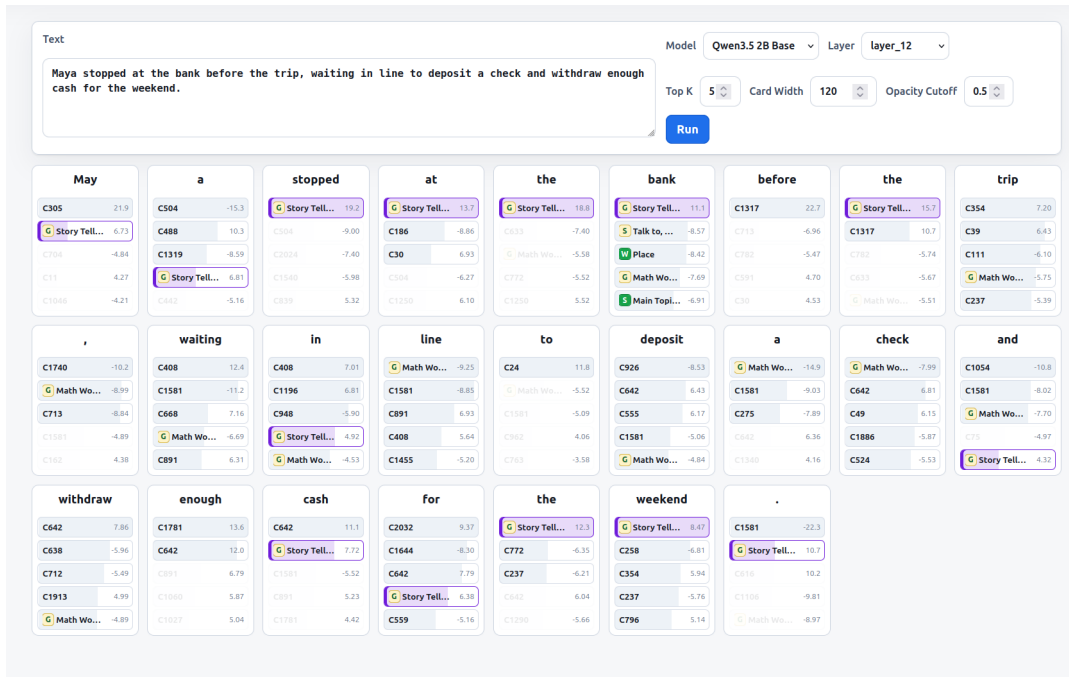


Figure 27: Explorer interface screenshots for Qwen 3.5 2B Base layer 12.